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Pathology slide automation system and method

Abstract

A slide carrier includes: a base support; and a slide platform having a surface that is parallel to a first plane defined by a first vector and a second vector, wherein a vector extending in a direction opposite to the direction of gravity is normal with respect to a second plane defined by a third vector and a fourth vector, an angle between the first vector and the third vector is greater than zero degrees and less than 90 degrees, and an angle between the second vector and the fourth vector is greater than zero degrees and less than 90 degrees.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a bypass continuation application of International Application PCT/US2023/017333, filed Apr. 3, 2023, which claims priority to and the benefit of U.S. Provisional Patent Application No. 63/327,744, filed on Apr. 5, 2022, and further claims priority to and the benefit of U.S. Provisional Patent Application No. 63/456,345, filed on Mar. 31, 2023, the entire contents of each of which are incorporated herein by reference.

FIELD

(1) One or more aspects of some embodiments according to the present disclosure relate to a pathology slide manufacturing system and method.

BACKGROUND

(2) In the field of pathology, the process of manufacturing individual slides having tissue or fluid samples may be complex and time consuming. Various steps and operations may be involved from when the sample is collected to when the final slide is ready to be reviewed by a pathologist. Additionally, when attempting to manufacture multiple slides, for multiple tissue or fluid samples, and from multiple patients, the process becomes even more complex.

(3) The above information disclosed in this Background section is only for enhancement of understanding of the background and therefore the information discussed in this Background section does not necessarily constitute prior art.

SUMMARY

(4) According to some embodiments of the present disclosure, a slide carrier includes: a base support; and a slide platform having a surface that is parallel to a first plane defined by a first vector and a second vector, wherein a vector extending in a direction opposite to the direction of gravity is normal with respect to a second plane defined by a third vector and a fourth vector, an angle between the first vector and the third vector is greater than zero degrees and less than 90 degrees, and an angle between the second vector and the fourth vector is greater than zero degrees and less than 90 degrees.

(5) According to some embodiments, the slide platform comprises a first long edge and a second long edge, wherein an elevation of the first long edge is higher than an elevation of the second long edge.

(6) According to some embodiments, the slide carrier further includes a barrier protruding from the slide platform at the second long edge.

(7) According to some embodiments, the slide carrier further includes a protrusion extending from the barrier.

(8) According to some embodiments, the slide platform comprises a first elongated support portion extending in a length direction of the slide platform and a second elongated support portion extending in the length direction of the slide platform, wherein the first elongated support portion and the second elongated support portion are separated from each other by a gap.

(9) According to some embodiments, the slide carrier further includes a crossmember support portion extending between and connecting the first elongated support portion and the second elongated support portion.

(10) According to some embodiments, the slide carrier further includes a first narrow support portion connected to a top of the first elongated support portion, wherein a top surface of the first

narrow support portion is coplanar with a top surface of the first elongated support portion; and a second narrow support portion connected to a top of the second elongated support portion, wherein a top surface of the second narrow support portion is coplanar with a top surface of the second elongated support portion, and a width between the first narrow support portion and the second narrow support portion is less than a width between the first elongated support portion and the second elongated support portion.

(11) According to some embodiments of the present disclosure, pathology slide manufacturing system includes: a mechanical arm; a gripper connected to the mechanical arm; and a slide carrier within reaching proximity of the gripper, the slide carrier comprising a slide platform having a surface that is parallel to a first plane defined by a first vector and a second vector, wherein a vector extending in a direction opposite to the direction of gravity is normal with respect to a second plane defined by a third vector and a fourth vector, an angle between the first vector and the third vector is greater than zero degrees and less than 90 degrees, and an angle between the second vector and the fourth vector is greater than zero degrees and less than 90 degrees.

(12) According to some embodiments, the slide platform comprises a first long edge and a second long edge, wherein an elevation of the first long edge is higher than an elevation of the second long edge.

(13) According to some embodiments, the pathology slide manufacturing system further includes a barrier protruding from the slide platform at the second long edge.

(14) According to some embodiments, the pathology slide manufacturing system further includes a protrusion extending from the barrier.

(15) According to some embodiments, the slide platform comprises a first elongated support portion extending in a length direction of the slide platform and a second elongated support portion extending in the length direction of the slide platform, wherein the first elongated support portion and the second elongated support portion are separated from each other by a gap.

(16) According to some embodiments, the pathology slide manufacturing system further includes a crossmember support portion extending between and connecting the first elongated support portion and the second elongated support portion.

(17) According to some embodiments, the pathology slide manufacturing system further includes a first narrow support portion connected to a top of the first elongated support portion, wherein a top surface of the first narrow support portion is coplanar with a top surface of the first elongated support portion; and a second narrow support portion connected to a top of the second elongated support portion, wherein a top surface of the second narrow support portion is coplanar with a top surface of the second elongated support portion, and a width between the first narrow support portion and the second narrow support portion is less than a width between the first elongated support portion and the second elongated support portion.

(18) According to some embodiments of the present disclosure, in a method of manufacturing a pathology slide in a slide processing system, the slide processing system includes: a mechanical arm; a gripper connected to the mechanical arm; and a slide carrier within reaching proximity of the gripper, and the method includes: gripping a slide with the gripper in a first orientation; placing, by the gripper, the slide on a slide platform of the slide carrier; allowing the slide to shift locations on the slide platform to register at a predetermined location on the slide platform; gripping the slide with the gripper in a second orientation corresponding to a component of the slide processing system; and placing, by the gripper, the slide in the component of the slide processing system using the gripper with the slide in the second orientation to enable a slide processing operation that corresponds to the component to be performed on the slide by the slide processing system.

(19) According to some embodiments, the slide platform has a surface that is parallel to a first plane defined by a first vector and a second vector, wherein a vector extending in a direction opposite to the direction of gravity is normal with respect to a second plane defined by a third vector and a fourth vector, an angle between the first vector and the third vector is greater than zero

degrees and less than 90 degrees, and an angle between the second vector and the fourth vector is greater than zero degrees and less than 90 degrees.

(20) According to some embodiments, the slide platform comprises a first long edge and a second long edge, wherein an elevation of the first long edge is higher than an elevation of the second long edge.

(21) According to some embodiments, the slide carrier comprises a barrier protruding from the slide platform at the second long edge, and the predetermined location corresponds to the barrier.

(22) According to some embodiments, a protrusion extends from the barrier.

(23) According to some embodiments, at least one of the first orientation or the second orientation involves a widthwise grip in which the gripper applies a compression force against opposing edges of the slide in a width direction.

(24) According to some embodiments, at least one of the first orientation or the second orientation involves a lengthwise grip in which the gripper applies a compression force against opposing edges of the slide in a length direction.

(25) According to some embodiments, at least one of the first orientation or the second orientation involves a thickness grip in which the gripper applies a compression force against opposing faces of the slide.

(26) According to some embodiments, the slide platform comprises a first elongated support portion extending in a length direction of the slide platform and a second elongated support portion extending in the length direction of the slide platform, wherein the first elongated support portion and the second elongated support portion are separated from each other by a gap.

(27) According to some embodiments, a crossmember support portion extends between and connects the first elongated support portion and the second elongated support portion.

(28) According to some embodiments, the slide platform further includes: a first narrow support portion connected to a top of the first elongated support portion, wherein a top surface of the first narrow support portion is coplanar with a top surface of the first elongated support portion; and a second narrow support portion connected to a top of the second elongated support portion, wherein a top surface of the second narrow support portion is coplanar with a top surface of the second elongated support portion, and a width between the first narrow support portion and the second narrow support portion is less than a width between the first elongated support portion and the second elongated support portion.

(29) According to some embodiments of the present disclosure, a gripper includes: a first gripper finger having a first groove and a second groove at an internal edge of the first gripper finger; and a second gripper finger having a first groove and a second groove at an internal edge of the second gripper finger, wherein the first groove of the first gripper finger faces the first groove of the second gripper finger such that the gripper is configured to apply a first compression force against length edges of a slide in a widthwise direction at the first grooves, and the second groove of the first gripper finger faces the second groove of the second gripper finger such that the gripper is configured to apply a second compression force against width edges of the slide in a lengthwise direction at the second grooves.

(30) According to some embodiments, a distance between the first groove of the first gripper finger and the first groove of the second gripper finger is less than a distance between the second groove of the first gripper finger and the second groove of the second gripper finger.

(31) According to some embodiments, the first groove of the first gripper finger is at a first portion of the first gripper finger that extends from an actuator in a direction parallel to a center line between the first and second gripper fingers.

(32) According to some embodiments, the second groove of the first gripper finger is at a second portion of the gripper finger that extends from the first portion in a direction that is at an angle less than 90 degrees and greater than zero degrees relative to the center line.

(33) According to some embodiments, the second groove of the first gripper finger is defined by a

space between: a protrusion extending from the first gripper finger toward a center line that bisects a space between the first and second gripper fingers; and a main body of the first gripper finger.

(34) According to some embodiments, the first gripper finger and the second gripper finger each comprise a protrusion at an end of the first gripper finger and the second gripper finger, respectively, which extend in a direction parallel to a center line that bisects the first and second gripper fingers, a distance between the protrusion of the first gripper finger and the protrusion of the second gripper finger is greater than a distance between the second groove of the first gripper finger and the second groove of the second gripper finger, and the gripper is configured to apply a third compression force against the width edges of the slide in the lengthwise direction.

(35) According to some embodiments, the first gripper finger comprises a third groove at an exterior edge of the first gripper finger, the third groove extending perpendicular to a center line that bisects a space between the first and second gripper fingers.

(36) According to some embodiments, the first gripper finger has a third groove extending in a direction parallel to the first groove of the first gripper finger, and the second gripper finger has a third groove facing the third groove of the first gripper finger and extending in the direction parallel to the first groove of the first gripper finger, wherein a distance between the third groove of the first gripper finger and the third groove of the second gripper finger is greater than a distance between the first groove of the first gripper finger and the first groove of the second gripper finger.

(37) According to some embodiments, the gripper is configured to apply a third compression force against the width edges of the slide in the lengthwise direction with an orientation in which the slide is rotated 90 degrees about an axis running parallel to a length of the slide compared to an orientation corresponding to a gripping position of the second grooves.

(38) According to some embodiments of the present disclosure, a pathology slide manufacturing system includes: a mechanical arm; and a gripper connected to the mechanical arm, the gripper comprising: a first gripper finger having a first groove and a second groove at an internal edge of the first gripper finger; and a second gripper finger having a first groove and a second groove at an internal edge of the second gripper finger, wherein the first groove of the first gripper finger faces the first groove of the second gripper finger such that the gripper is configured to apply a compression force against length edges of a slide in a widthwise direction at the first grooves, and the second groove of the first gripper finger faces the second groove of the second gripper finger such that the gripper is configured to apply a compression force against width edges of the slide in a lengthwise direction at the second grooves.

(39) According to some embodiments, a distance between the first groove of the first gripper finger and the first groove of the second gripper finger is less than a distance between the second groove of the first gripper finger and the second groove of the second gripper finger.

(40) According to some embodiments, the first groove of the first gripper finger is at a first portion of the first gripper finger that extends from an actuator in a direction parallel to a center line between the first and second gripper fingers.

(41) According to some embodiments, the second groove of the first gripper finger is at a second portion of the gripper finger that extends from the first portion in a direction that is at an angle less than 90 degrees and greater than zero degrees relative to the center line.

(42) According to some embodiments, the second groove of the first gripper finger is defined by a space between: a protrusion extending from the first gripper finger toward a center line that bisects a space between the first and second gripper fingers; and a main body of the first gripper finger.

(43) According to some embodiments, the first gripper finger and the second gripper finger each comprise a protrusion at an end of the first gripper finger and the second gripper finger, respectively, which extend in a direction parallel to a center line that bisects the first and second gripper fingers, a distance between the protrusion of the first gripper finger and the protrusion of the second gripper finger is greater than a distance between the second groove of the first gripper finger and the second groove of the second gripper finger, and the gripper is configured to apply a

third compression force against the width edges of the slide in the lengthwise direction.

(44) According to some embodiments, the first gripper finger comprises a third groove at an exterior edge of the first gripper finger, the third groove extending perpendicular to a center line that bisects a space between the first and second gripper fingers.

(45) According to some embodiments, the first gripper finger has a third groove extending in a direction parallel to the first groove of the first gripper finger, and the second gripper finger has a third groove facing the third groove of the first gripper finger and extending in the direction parallel to the first groove of the first gripper finger, wherein a distance between the third groove of the first gripper finger and the third groove of the second gripper finger is greater than a distance between the first groove of the first gripper finger and the first groove of the second gripper finger.

(46) According to some embodiments, the gripper is configured to apply a third compression force against the width edges of the slide in the lengthwise direction with an orientation in which the slide is rotated 90 degrees about an axis running parallel to a length of the slide compared to an orientation corresponding to a gripping position of the second grooves.

(47) According to some embodiments of the present disclosure, in a method of manufacturing a slide in a slide processing system, the slide processing system includes: a mechanical arm; and a gripper connected to the mechanical arm, the gripper comprising: a first gripper finger having a first groove and a second groove at an internal edge of the first gripper finger; and a second gripper finger having a first groove and a second groove at an internal edge of the second gripper finger, wherein the first groove of the first gripper finger faces the first groove of the second gripper finger such that the gripper is configured to apply a first compression force against length edges of a slide in a widthwise direction at the first grooves, and the second groove of the first gripper finger faces the second groove of the second gripper finger such that the gripper is configured to apply a second compression force against width edges of the slide in a lengthwise direction at the second grooves, and the method includes: gripping the slide in a first orientation using the first groove of the first gripper finger and the first groove of the second gripper finger; placing the slide in a first component of the slide processing system in the first orientation; gripping the slide in a second orientation using the second groove of the first gripper finger and the second groove of the second gripper finger; and placing the slide in a second component of the slide processing system in the second orientation.

(48) According to some embodiments, a distance between the first groove of the first gripper finger and the first groove of the second gripper finger is less than a distance between the second groove of the first gripper finger and the second groove of the second gripper finger.

(49) According to some embodiments, the first groove of the first gripper finger is at a first portion of the first gripper finger that extends from an actuator in a direction parallel to a center line between the first and second gripper fingers.

(50) According to some embodiments, the second groove of the first gripper finger is at a second portion of the gripper finger that extends from the first portion in a direction that is at an angle less than 90 degrees and greater than zero degrees relative to the center line.

(51) According to some embodiments, the second groove of the first gripper finger is defined by a space between: a protrusion extending from the first gripper finger toward a center line that bisects a space between the first and second gripper fingers; and a main body of the first gripper finger.

(52) According to some embodiments, the first gripper finger comprises a protrusion at an end of the gripper finger and extending in a direction parallel to a center line that bisects the first and second gripper fingers.

(53) According to some embodiments, wherein the first gripper finger comprises a third groove at an exterior edge of the first gripper finger, the third groove extending perpendicular to a center line that bisects a space between the first and second gripper fingers.

(54) According to some embodiments, the method further includes: aligning the third groove with an edge of the slide; and applying a horizontal or vertical force, relative to a face of the slide,

against the edge of the slide.

(55) According to some embodiments, the first gripper finger has a third groove extending in a direction parallel to the first groove of the first gripper finger, and the second gripper finger has a third groove facing the third groove of the first gripper finger and extending in the direction parallel to the first groove of the first gripper finger, wherein a distance between the third groove of the first gripper finger and the third groove of the second gripper finger is greater than a distance between the first groove of the first gripper finger and the first groove of the second gripper finger.

(56) According to some embodiments, the method further includes applying a third compression force against the width edges of the slide in the lengthwise direction with an orientation in which the slide is rotated 90 degrees about an axis running parallel to a length of the slide compared to an orientation corresponding to a gripping position of the second grooves.

(57) According to some embodiments of the present disclosure, a slide carrier cassette system includes: a slide carrier cassette having a main body; and a plurality of slots in the main body and aligned along an alignment direction, wherein each slot has an opening at a top side of the main body and is configured to receive a slide, and each slot is at an angle less than 90 degrees and greater than 0 degrees relative to the alignment direction.

(58) According to some embodiments, each slot comprises a first portion having a thickness corresponding to a thickness of the slide, and a second portion having a thickness that is greater than the thickness of the first portion.

(59) According to some embodiments, the slide carrier cassette system further includes: a spring at an exterior side of the main body and configured to engage with a cavity of a slide carrier cassette queue to retain the slide carrier cassette in the cavity.

(60) According to some embodiments, the slide carrier cassette system further includes: a storage queue having a plurality of cavities having a shape such that each cavity is configured to receive a slide carrier cassette.

(61) According to some embodiments, the cavities are arranged in a matrix of rows and columns.

(62) According to some embodiments, the queue has a cylindrical shape and the cavities are arranged around a central axis.

(63) According to some embodiments, each cavity is rotated clockwise or counterclockwise relative to a vertical line at an angle less than 90 degrees and greater than 0 degrees.

(64) According to some embodiments, the slide carrier cassette further comprises a tab handle at a top side of the slide carrier cassette adjacent to the openings of the slots.

(65) According to some embodiments of the present disclosure, a pathology slide manufacturing system includes: a mechanical arm; a gripper connected to the mechanical arm; and a slide carrier cassette system within reaching proximity of the gripper, the slide carrier cassette system comprising: a slide carrier cassette having a main body; and a plurality of slots in the main body and aligned along an alignment direction, wherein each slot has an opening at a top side of the main body and is configured to receive a slide, and each slot is at an angle less than 90 degrees and greater than 0 degrees relative to the alignment direction.

(66) According to some embodiments, each slot comprises a first portion having a thickness corresponding to a thickness of the slide, and a second portion having a thickness that is greater than the thickness of the first portion.

(67) According to some embodiments, the pathology slide manufacturing system further includes a spring at an exterior side of the main body and configured to engage with a cavity of a slide carrier cassette queue to retain the slide carrier cassette in the cavity.

(68) According to some embodiments, the pathology slide manufacturing system further includes a storage queue having a plurality of cavities having a shape such that each cavity is configured to receive a slide carrier cassette.

(69) According to some embodiments, the cavities are arranged in a matrix of rows and columns.

(70) According to some embodiments, the queue has a cylindrical shape and the cavities are

arranged around a central axis.

(71) According to some embodiments, each cavity is rotated clockwise or counterclockwise relative to a vertical line at an angle less than 90 degrees and greater than 0 degrees.

(72) According to some embodiments, the slide carrier cassette further comprises a tab handle at a top side of the slide carrier cassette adjacent to the openings of the slots.

(73) According to some embodiments of the present disclosure, in a method of manufacturing a pathology slide in a slide processing system, the slide processing system comprising: a mechanical arm; and a gripper connected to the mechanical arm; and a slide carrier cassette within reaching proximity of the gripper, and the method includes: gripping a slide with the gripper; inserting, with the gripper, the slide into a slot of the slide carrier cassette; placing, with the gripper, the slide carrier cassette into a queue having a cavity corresponding to a shape of the slide carrier cassette; removing, with the gripper, the slide from the slide carrier cassette; and placing, with the gripper, the slide into a component of the slide processing system to enable a slide manufacturing operation to be performed on the slide.

(74) According to some embodiments, the slide carrier cassette comprises: a main body; and a plurality of slots in the main body and aligned along an alignment direction, wherein each slot has an opening at a top side of the main body and is configured to receive a slide, and each slot is at an angle less than 90 degrees and greater than 0 degrees relative to the alignment direction.

(75) According to some embodiments, each slot comprises a first portion having a thickness corresponding to a thickness of the slide, and a second portion having a thickness that is greater than the thickness of the first portion.

(76) According to some embodiments, the slide carrier cassette further includes a spring at an exterior side of the main body and configured to engage with a cavity of a slide carrier cassette queue to retain the slide carrier cassette in the cavity.

(77) According to some embodiments, the cavities are arranged in a matrix of rows and columns.

(78) According to some embodiments, the queue has a cylindrical shape and the cavities are arranged around a central axis.

(79) According to some embodiments, each cavity is rotated clockwise or counterclockwise relative to a vertical line at an angle less than 90 degrees and greater than 0 degrees.

(80) According to some embodiments, the slide carrier cassette further includes a tab handle at a top side of the slide carrier cassette adjacent to the openings of the slots.

(81) According to some embodiments of the present disclosure, a slide processing system includes: a mechanical arm; a slide portal comprising a plurality of portal slots; a processor in communication with the mechanical arm and the slide portal; a memory in communication with the processor, the memory storing instructions that, when executed by the processor, cause the processor to: receive slide information; identify a portal slot from among the portal slots for receiving slides corresponding to the slide information; transmit a signal to the mechanical arm to insert a carriage into the portal slot, wherein the carriage has an internal cavity that corresponds to the slide information; transmit a signal to the mechanical arm to retrieve the carriage from the portal slot; and transmit a signal to the mechanical arm to remove the slides from the carriage and store the slides in a storage location.

(82) According to some embodiments, the slide information comprises information indicating whether the slide is located in a slide tray or a slide rack and information indicating a form factor of the slide tray or slide rack.

(83) According to some embodiments, the instructions further cause the processor to transmit a signal indicating a slide tray or slide rack is inserted into the carriage prior to transmission of the signal to the mechanical arm to retrieve the carriage from the portal slot.

(84) According to some embodiments, the instructions further cause the processor to transmit a signal to the mechanical arm to transfer the slides to a slide carrier cassette and to transfer the slide carrier cassette to a slide carrier queue as the storage location.

(85) According to some embodiments, the instructions further cause the processor to transmit a signal to the mechanical arm to return the carriage to the portal slot after all of the slides are removed.

(86) According to some embodiments, the slide processing system further comprises a user interface comprising a display panel, and the instructions further cause the processor to receive the slide information as user input at the user interface.

(87) According to some embodiments, the slide processing system further includes a slide position sensor, wherein the instructions further cause the processor to transmit a signal to the slide position sensor to detect a slide position of the slides before removing the slides from the carriage.

(88) According to some embodiments, the instructions further cause the processor to: transmit a signal to the slide position sensor to initiate slide position detection; transmit a signal to the slide position sensor to capture one or more images; identify a position of a slide from among the slides based on the one or more images; and transmit a signal to the mechanical arm to remove the slide after the position of the slide is identified.

(89) According to some embodiments of the present disclosure, in a method of processing slides, the method includes: receiving, by a processor, slide information; identifying, by the processor, a portal slot from among the portal slots for receiving a slide corresponding to the slide information; transmitting, by the processor, a signal to the mechanical arm to insert a carriage into the portal slot, wherein the carriage has an internal cavity that corresponds to the slide information; transmitting, by the processor, a signal to the mechanical arm to retrieve the carriage from the portal slot; and transmitting, by the processor, a signal to the mechanical arm to remove the slides from the carriage and store the slides in a storage location.

(90) According to some embodiments, the slide information comprises information indicating whether the slide is located in a slide tray or a slide rack and information indicating a form factor of the slide tray or slide rack.

(91) According to some embodiments, the method further includes transmitting, by the processor, a signal indicating a slide tray or slide rack is inserted into the carriage prior to transmission of the signal to the mechanical arm to retrieve the carriage from the portal slot.

(92) According to some embodiments, the method further includes transmitting, by the processor, a signal to the mechanical arm to transfer the slides to a slide carrier cassette and to transfer the slide carrier cassette to a slide carrier queue as the storage location.

(93) According to some embodiments, the method further includes transmitting, by the processor, a signal to the mechanical arm to return the carriage to the portal slot after all of the slides are removed.

(94) According to some embodiments, the method further includes receiving, by the processor, the slide information as user input at a user interface, the user interface comprising a display panel.

(95) According to some embodiments, the method further includes transmitting, by the processor, a signal to a slide position sensor to detect a slide position of the slides before removing the slides from the carriage.

(96) According to some embodiments, the method further includes: transmitting, by the processor, a signal to the slide position sensor to initiate slide position detection; transmitting, by the processor, a signal to the slide position sensor to capture one or more images; identifying, by the processor, a position of a slide from among the slides based on the one or more images; and transmitting, by the processor, a signal to the mechanical arm to remove the slide after the position of the slide is identified.

(97) According to some embodiments of the present disclosure, a slide processing system includes: a mechanical arm; a slide portal comprising a plurality of portal slots; a processor in communication with the mechanical arm and the slide portal; a memory in communication with the processor, the memory storing instructions that, when executed by the processor, cause the processor to: receive slide information; identify a portal slot from among the portal slots for

depositing slides corresponding to the slide information; transmit a signal to the mechanical arm to retrieve the slides corresponding to the slide information; transmit a signal to the mechanical arm to insert the slides corresponding to the slide information into a tray or rack; and transmit a signal to the mechanical arm to transfer the tray or rack having the slides to the portal slot.

(98) According to some embodiments, the instructions further cause the processor to transmit a signal to a user interface, the user interface comprising a display panel, to display a prompt to insert the tray or rack corresponding to the slide information into the portal slot.

(99) According to some embodiments, the instructions further cause the processor to transmit a signal to the mechanical arm to remove the tray or rack from the portal slot and transfer the tray or rack to a nest.

(100) According to some embodiments, the instructions further cause the processor to transmit a signal to the mechanical arm to insert the slides corresponding to the slide information into the tray or rack in the nest.

(101) According to some embodiments, the instructions further cause the processor to identify the slides in a slide storage system corresponding to the slide information.

(102) According to some embodiments, the slide processing system further includes a user interface comprising a display panel, wherein the instructions further cause the processor to receive the slide information as user input at the user interface.

(103) According to some embodiments of the present disclosure, in a method of processing slides, the method includes: receiving, by a processor, slide information; identifying, by the processor, a portal slot from among the portal slots for depositing slides corresponding to the slide information; transmitting, by the processor, a signal to the mechanical arm to retrieve the slides corresponding to the slide information; transmitting, by the processor, a signal to the mechanical arm to insert the slides corresponding to the slide information into a tray or rack; and transmitting, by the processor, a signal to the mechanical arm to transfer the tray or rack having the slides to the portal slot.

(104) According to some embodiments, the method further includes: transmitting, by the processor, a signal to a user interface, the user interface comprising a display panel, to display a prompt to insert the tray or rack corresponding to the slide information into the portal slot.

(105) According to some embodiments, the method further includes: transmitting, by the processor, a signal to the mechanical arm to remove the tray or rack from the portal slot and transfer the tray or rack to a nest.

(106) According to some embodiments, the method further includes: transmitting, by the processor, a signal to the mechanical arm to insert the slides corresponding to the slide information into the tray or rack in the nest.

(107) According to some embodiments, the method further includes: identifying, by the processor, the slides in a slide storage system corresponding to the slide information.

(108) According to some embodiments, the method further includes: receiving, by the processor, the slide information as user input at a user interface.

(109) According to some embodiments of the present disclosure, a gripper includes: an actuator; a first gripper finger coupled to the actuator, the first gripper finger having a first groove at an internal edge of a main body of the first gripper finger and a first protrusion extending from the main body of the first gripper finger with a second groove between the first protrusion and the main body of the first gripper finger; a second gripper finger coupled to the actuator, the second gripper finger having a second groove at an internal edge of a main body of the second gripper finger and a second protrusion extending from the main body of the second gripper finger with a second groove between the second protrusion and the main body of the second gripper finger, wherein the first gripper finger and the second gripper finger are configured to apply a first compression force against length edges of a slide in a widthwise direction at the first grooves in a first gripping mode, and the first gripper finger and the second gripper finger are configured to apply a second compression force against width edges of the slide in a lengthwise direction at the second grooves

in a second gripping mode.

(110) According to some embodiments, the first gripper finger and the second gripper finger are equidistant from a center line between the first gripper finger and the second gripper finger and are configured to move toward and away from the center line in coordination with each other.

(111) According to some embodiments, a distance between the first groove of the first and second gripper fingers is less than a distance between the second groove of the first and second gripper fingers.

(112) According to some embodiments, the gripper further includes a gripper bracket coupled between the actuator and the first gripper finger, wherein the first gripper finger is configured to be connected and disconnected from the bracket.

(113) According to some embodiments of the present disclosure, a slide processing system includes: a slide carriage having a first channel on a first side of the slide carriage and a second channel of the slide carriage; an actuator; a first gripper rod coupled to the actuator; and a second gripper rod coupled to the actuator, wherein the first gripper rod and the second gripper rod are configured to fit within the first channel and the second channel respectively.

(114) According to some embodiments, the first gripper rod comprises a plurality of segments separated by a flared portion.

(115) According to some embodiments, the flared portion comprises an O-ring.

(116) According to some embodiments, the slide processing system further includes a bracket coupled between the actuator and the first gripper rod.

(117) According to some embodiments, the slide carriage is a tray carriage.

(118) According to some embodiments, the slide carriage is a rack carriage.

(119) According to some embodiments, the slide carriage comprises a plurality of slide carriages, wherein a first slide carriage from among the slide carriages has an internal cavity with a first interior form factor corresponding to a first type of slide tray and a second slide carriage from among the slide carriages has an internal cavity with a second interior form factor corresponding to a second type of slide tray, wherein the first interior form factor is different from the second interior form factor, and wherein the first slide carriage and the second slide carriage have a same exterior form factor.

(120) According to some embodiments, the slide carriage comprises a plurality of slide carriages, wherein a first slide carriage from among the slide carriages has an internal cavity with a first interior form factor corresponding to a first type of slide rack and a second slide carriage from among the slide carriages has an internal cavity with a second interior form factor corresponding to a second type of slide rack, wherein the first interior form factor is different from the second interior form factor, and wherein the first slide carriage and the second slide carriage have a same exterior form factor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting and non-exhaustive embodiments according to the present disclosure are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified.

(2) FIG. 1 is a flow diagram illustrating various operations that may occur in a pathology context or pathology environment, according to some embodiments;

(3) FIG. 2A illustrates slide processing or manufacturing environment, according to some embodiments;

(4) FIG. 2B illustrates an example of a slide after being manufactured, according to some embodiments;

(5) FIG. 3A is a right front perspective view of a slide carrier, according to some embodiments;

(6) FIG. 3B is a front view of the slide carrier, according to some embodiments;

(7) FIG. 3C is a right side view of the slide carrier, according to some embodiments;

(8) FIG. 3D is a left side view of the slide carrier, according to some embodiments;

(9) FIG. 3E is a left front perspective view of the slide carrier, according to some embodiments;

(10) FIG. 3F is a right rear perspective view of the slide carrier, according to some embodiments;

(11) FIG. 3G is a rear view of the slide carrier, according to some embodiments;

(12) FIG. 3H is a top view of the slide carrier, according to some embodiments;

(13) FIG. 3I is a bottom view of the slide carrier, according to some embodiments;

(14) FIG. 3J is a flow diagram illustrating a method of manufacturing a pathology slide in a slide processing system, according to some embodiments;

(15) FIG. 4A illustrates a gripper holding a slide in a first gripping mode, according to some embodiments;

(16) FIG. 4B illustrates the gripper holding a slide in a second gripping mode, according to some embodiments;

(17) FIG. 4C illustrates the gripper holding a slide in a third gripping mode, according to some embodiments;

(18) FIG. 4D illustrates the gripper holding a slide in a fourth gripping mode, according to some embodiments;

(19) FIG. 4E illustrates the gripper manipulating a slide using an exterior groove, according to some embodiments;

(20) FIG. 4F is a top front perspective view of a gripper finger, according to some embodiments;

(21) FIG. 4G is a top rear perspective view of a gripper finger, according to some embodiments;

(22) FIG. 4H is a bottom view of a gripper finger, according to some embodiments;

(23) FIG. 4I is a close-up view of a section A of FIG. H, according to some embodiments;

(24) FIG. 4J is a close-up view of a section B of FIG. H, according to some embodiments;

(25) FIG. 4K is a left side view of a gripper finger, according to some embodiments;

(26) FIG. 4L is a close-up view of a section C of FIG. 4K, according to some embodiments;

(27) FIG. 4M is a flow diagram illustrating a method of manufacturing a slide in a slide processing system, according to some embodiments;

(28) FIG. 5A is a front perspective view of a slide carrier cassette, according to some embodiments;

(29) FIG. 5B is a rear perspective view of a slide carrier cassette, according to some embodiments;

(30) FIG. 5C is a front view of a slide carrier cassette, according to some embodiments;

(31) FIG. 5D is a cross-sectional view taken along the line A-A of FIG. 5C, according to some embodiments;

(32) FIG. 5E is a close-up view of the section B of FIG. 5D, according to some embodiments;

(33) FIG. 5F is shows an example of a queue or storage rack for a plurality of slide carrier cassettes, according to some embodiments;

(34) FIG. 5G shows an example of a queue or storage rack with a cylindrical shape, according to some embodiments;

(35) FIG. 5H shows an example of a queue or storage rack in a stacked configuration according to some embodiments;

(36) FIG. 5I shows an example of a slide processing environment, according to some embodiments;

(37) FIG. 5J is a flow diagram illustrating a method of manufacturing a slide in a slide processing system, according to some embodiments;

(38) FIG. 6A illustrates aspects of an example automated histology storage system and environment, according to some embodiments;

(39) FIG. 6B illustrates an example automated histology storage system, according to some embodiments;

(40) FIG. 6C shows further details of an example automated histology storage system, according to some embodiments;

(41) FIG. 6D shows an example top view of the automated histology storage system, according to some embodiments;

(42) FIG. 7A shows an example of a variety of slide racks, according to some embodiments,

(43) FIG. 7B shows an example of a variety of slide trays, according to some embodiments;

(44) FIG. 7C shows example input or output portals, according to some embodiments;

(45) FIG. 7D shows further details of an example tray portal according to some embodiments;

(46) FIG. 7E shows further details of internal components of a tray portal according to some embodiments;

(47) FIG. 7F shows further details of example tray carriages, according to some embodiments;

(48) FIG. 7G shows a bottom view of a portion of a portal slot of a tray portal, according to some embodiments;

(49) FIG. 7H shows a close-up view of a rail of an example tray carriage, according to some embodiments;

(50) FIG. 7I shows a close-up view of a portion of the internal structure of a tray portal slot when a carrier tray is in an extended or pulled-out position, according to some embodiments;

(51) FIG. 7J shows a close-up view of the portion of the internal structure of the portal slot when the carrier tray is in a retracted or pushed-in position, according to some embodiments;

(52) FIG. 7K illustrates further details of internal components of a rack portal, according to some embodiments;

(53) FIG. 7L shows further details of example rack carriages, according to some embodiments;

(54) FIG. 7M shows an example rack carriage nest, according to some embodiments;

(55) FIG. 7N shows an example tray carriage nest, according to some embodiments;

(56) FIG. 7O shows a side view of an example rack nest, according to some embodiments;

(57) FIG. 7P shows a perspective view of the example rack nest, according to some embodiments;

(58) FIG. 8A shows an example slide position detection system, according to some embodiments;

(59) FIG. 8B shows an example image of a first slide position in a slide position detection system, according to some embodiments;

(60) FIG. 8C shows an example image of a second slide position in a slide position detection system, according to some embodiments;

(61) FIG. 8D illustrates an example method to determine one or more slide positions, according to some embodiments;

(62) FIG. 8E illustrates an example method to determine one or more slide positions, according to some embodiments;

(63) FIG. 9A illustrates an example of a gripper holding a slide in a first gripping mode, according to some embodiments;

(64) FIG. 9B illustrates an example of a gripper holding a slide in a second gripping mode, according to some embodiment;

(65) FIG. 9C illustrates further details of an example gripper, according to some embodiments;

(66) FIG. 9D illustrates further details of an internal edge of a gripper finger, according to some embodiments;

(67) FIG. 9E illustrates further details of a gripper finger, according to some embodiments;

(68) FIG. 10A illustrates an example of a gripper, according to some embodiments;

(69) FIG. 10B illustrates an example gripper finger rod, according to some embodiments;

(70) FIG. 10C illustrates example gripping channels of a rack carriage, according to some embodiments;

(71) FIG. 10D illustrates example gripping channels of a tray carriage, according to some embodiments;

(72) FIG. 10E illustrates an example of a gripper holding a rack carriage, according to some

embodiments;

(73) FIG. 10F illustrates an example of a gripper holding a tray carriage, according to some embodiments;

(74) FIG. 11A illustrates an example of a gripper, according to some embodiments;

(75) FIG. 11B illustrates a rear view of a gripper, according to some embodiments;

(76) FIG. 12A illustrates an example process for an automated histology storage system to receive slides from a user or technician, according to some embodiments; and

(77) FIG. 12B illustrates an example process for an automated histology storage system to retrieve slides from storage and transfer them to a user or technician, according to some embodiments.

DETAILED DESCRIPTION

(78) Hereinafter, aspects of some example embodiments will be described in more detail with reference to the accompanying drawings, in which like reference numbers refer to like elements throughout. The present invention, however, may be embodied in various different forms, and should not be construed as being limited to only the illustrated embodiments herein. Rather, these embodiments are provided as examples so that this disclosure will be thorough and complete, and will fully convey the aspects and features of the present invention to those skilled in the art. Accordingly, processes, elements, and techniques that are not necessary to those having ordinary skill in the art for a complete understanding of the aspects and features of the present invention may not be described. Unless otherwise noted, like reference numerals denote like elements throughout the attached drawings and the written description, and thus, descriptions thereof will not be repeated. In the drawings, the relative sizes of elements, layers, and regions may be exaggerated for clarity.

(79) It will be understood that, although the terms “first,” “second,” “third,” etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section described below could be termed a second element, component, region, layer or section, without departing from the spirit and scope of the present invention.

(80) Spatially relative terms, such as “beneath,” “below,” “lower,” “under,” “above,” “upper,” and the like, may be used herein for ease of explanation to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or in operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” or “under” other elements or features would then be oriented “above” the other elements or features. Thus, the example terms “below” and “under” can encompass both an orientation of above and below. The device may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein should be interpreted accordingly.

(81) It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it can be directly on, connected to, or coupled to the other element or layer, or one or more intervening elements or layers may be present. In addition, it will also be understood that when an element or layer is referred to as being “between” two elements or layers, it can be the only element or layer between the two elements or layers, or one or more intervening elements or layers may also be present.

(82) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present invention. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and “including,” when used in this specification, specify the presence of the stated features, integers, steps,

operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

(83) As used herein, the term “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent deviations in measured or calculated values that would be recognized by those of ordinary skill in the art.

Further, the use of “may” when describing embodiments of the present invention refers to “one or more embodiments of the present invention.” As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. Also, the term “exemplary” is intended to refer to an example or illustration.

(84) Pathology is the medical discipline, which attempts to facilitate the diagnosis and treatment of diseases, by studying tissue, cell, and fluid samples of patients. In many applications, tissue samples may be collected from patients, and processed into a form that can be analyzed by physicians (e.g., Pathologists), often under magnification, by physicians to diagnose and characterize relevant medical conditions based on the tissue sample.

(85) FIG. 1 is a flow diagram illustrating various operations that may occur in a pathology environment or pathology system **100**. For example, when a treating physician or medical provider identifies a patient for whom an analysis of a tissue or fluid sample may be beneficial for diagnosing or treating a medical condition, a tissue or fluid sample may be collected at operation **102**. The patient's identity may be collected and matched with the patient's sample, and the sample may be placed in a sterile container and/or collection medium for further processing.

(86) The sample may then be transported to a pathology accessioning laboratory at operation **104**, where the sample may be received, sorted, organized, and labeled along with other samples from other patients, for further processing.

(87) At operation **106**, the sample may be further processed as part of a grossing operation. For example, an individual tissue sample or specimen may be sliced into smaller sections for embedding and subsequent cutting for assembly onto slides.

(88) Then, at operation **108**, the sample or specimen may be mounted or deposited on one or more glass slides. The preparation of slides may involve applying one or more reagents or stains to the sample, for example, in order to improve the visibility of, or contrast between, different parts of the sample.

(89) In some instances, at operation **110**, several slides, either during the reagent or staining processing, or after the processing is completed, may be assembled or collected in a case or folio. The case may, for example, be carefully labeled with the individual patient's identifying information.

(90) Between each of the operations **102** and **110**, at operation **112**, the sample, specimen, slide(s), may be transported within the medical facility, or between medical facilities (e.g., between a physician's office and a laboratory), or may be stored between processing operations.

(91) Once the processing of the samples and slides is completed, and a pathologist is ready to review the sample, the slides and/or the case(s) holding multiple slides corresponding to the patient may again be transported, at operation **112**, to the pathologist. At operation **114**, the pathologist may review the slides, for example, under magnification using a microscope. An individual slide may be placed under the objective lens of the microscope, and the microscope and the slide may be manipulated and adjusted as the pathologist reviews the tissue or fluid.

(92) Once the pathologist case completed the review of the slide, the pathologist may attempt, at operation **116**, to form a medical opinion or diagnosis. Meanwhile, the sample or slides may once again be transported, at operation **112**, to a longer term storage facility. In some instances, the sample or slides may be again transported, either before or after some storage period, to other

physicians for further analysis, second opinions, and the like.

(93) Thus, as can be appreciated based on the pathology process flow described above, the pathology system **100** involves a wide variety of processing operations. At each operation, there is an opportunity for tissue samples to be damaged or contaminated, or for information to be lost. Additionally, the processing of individual slides can be quite time consuming and complicated, and it can be difficult for laboratories to manage processing a high volume of patient samples simultaneously. Often times, there are also time constraints, since earlier diagnosis of disease may lead to better patient outcomes, yet each operation in the pathology system **100** presents an opportunity for delay.

(94) Thus, embodiments according to the present disclosure may improve the efficiency and productivity of the pathology system **100**.

(95) FIG. 2A illustrates slide processing or manufacturing system **200**. According to some embodiments the slide processing system **200** may exist as part of the pathology system **100** illustrated in FIG. 1 (for example, as part of operations **108** and **110**).

(96) As shown in FIG. 2A, the slide processing system **200** may include one or more mechanical arms **202**. The mechanical arm **202** may be utilized to transfer a slide **204**, having a specimen **206** deposited thereon, from a slide carrier **208** to one or more slide processing stations **210a-210e**. Although FIG. 2A illustrates the slide processing system **200** includes five slide processing stations **210a-210e**, embodiments according to the present disclosure are not limited thereto, and the slide processing environment may include fewer slide processing stations or additional slide processing stations without departing from the spirit and scope of embodiments according to the present disclosure. Each of the slide processing stations **210a-210e** may be a mechanical apparatus configured to perform one or more specific processing operations during a process of manufacturing a slide. For example, the slide processing stations **210a-210e** may be configured to apply different and unique reagents or stains to the specimen **206**.

(97) In some embodiments, the slide carrier **208** may operate as a temporary platform or storage area between processing operations and/or between transfers from one of the slide processing stations **210a-210e** to another one of the slide processing stations **210a-210e** and may be configured to accept one or more slides at a time.

(98) Additionally, various other slide carriers **214a-214b** may be utilized in one or more processing operations. Although two slide carriers **214a-214b** are illustrated in FIG. 2A for the purpose of illustration, embodiments according to the present disclosure are not limited thereto, and the number and types of slide carriers that are utilized in any given manufacturing process may vary according to various embodiments depending on the slide processing stations **210a-210e** and the particular processing operations involved. For example, in some instances, different slide processing stations **210a-210e** may be designed to accommodate slide carriers with specific shapes or designs that are a function of the processing operation to be performed, the size and shape of the slide processing station, and the chemicals or treatments to be applied to the slide.

(99) As discussed above, between each processing operation, the mechanical arm **202** may be configured to transfer the slide **204** between different components within the slide processing system **200**. For example, the mechanical arm **202** may be configured to pick up the slide **204** from the slide carrier **208**, by using a gripper **216**, and transfer the slide **204** directly to one of the slide processing stations **210a-210e** to be inserted therein. The corresponding slide processing station **210a-210e** may then perform a processing operation on the slide **204** (for example, by applying a reagent or stain to the specimen **206**). After the processing operation is completed, the mechanical arm **202** may remove the slide **204** from the slide processing station **210a-210e**, and transfer the slide back to the slide carrier **208**, or transfer the slide **204** to another one of the slide processing stations **210a-210e** or to one of the slide carriers **214a-214b**. The slide manufacturing process may continue on in this manner, with the mechanical arm **202** transferring the slide **204** between different components or stations within the slide processing system **200**, until all of the

manufacturing operations have been completed and the slide is ready to be stored or transferred out of the slide processing system **200** for subsequent pathology operations.

(100) In order to physically pick up, maneuver, and/or manipulate the slide **204** during the various transferring operations, the mechanical arm **202** may include one or more grippers **216**. Although FIG. 2A illustrates two opposing gripper fingers **216a** and **216b** (which may be collectively referred to as a gripper **216** herein), embodiments according to the present invention are not limited thereto, and some embodiments may include various gripper configurations or designs. Further details of an example structure of the gripper **216** according to some embodiments is illustrated and described in more detail below.

(101) According to some embodiments, the mechanical arm **202** may be in electronic communication with a controller system **220** that is configured to control the movement and operation of the mechanical arm **202** and the gripper **216**, in order to move the slide **204** between different locations and components within the slide processing system **200**.

(102) The controller system **220** may include a processor **222** and a memory **224**, where the memory **224** stores instructions thereon, which, when executed by the processor **222**, cause the processor **222** to send and receive electronic (e.g., data) signals to and from the mechanical arm **202** and the gripper **216**, in order to control the movement and operation of the mechanical arm **202** (e.g., one or more actuators and motors of the mechanical arm **202**) and the gripper **216** (e.g., one or more gripper fingers of the gripper **216**). The instructions stored in the memory **224** may include, for example, instructions corresponding to one or more slide processing operations described herein, which may include, for example, placing a slide into or onto a slide carrier or processing machine or station, adjusting the gripping orientation of the gripper, and any other operations involved with manipulating and maneuvering a slide as part of the slide processing system **200**, in order to manufacture a pathology slide, as described in more detail below.

(103) FIG. 2B illustrates an example of a slide after being manufactured according to some embodiments. In some embodiments, the slide **204** includes a substrate **230** that may be formed of any suitable transparent material, such as glass. The sample or specimen (e.g., a slice of tissue, bodily fluid, cavity swap, etc.) **206** may be deposited on a front face of the slide **204**, which is opposite a rear face of the slide **204**. After depositing the specimen **206** onto the front face of the slide **204**, the specimen may be treated with various manufacturing processes or operations, for example, by applying one or more reagents and/or stains to the specimen **206**. A slip **232** may be placed over the specimen **206** to protect the integrity of the specimen **206** and prevent or reduce physical impact or contact with the specimen **206**, and prevent or reduce contamination by foreign contaminants. Hereinafter, a slide having a tissue sample or specimen with a slip covering the specimen may be referred to as a “coverslipped” slide.

(104) Additionally, a label **234** may be affixed to the front face of the slide at a location adjacent to the specimen and the slip. The label **234** may include various information that includes unique identifying information about the slide, to enable the content of the specimen, the patient, etc., to be identified. For example, the label **234** may include a barcode or other alphanumeric characters operating as an identification for the slide.

(105) The slip **232** may be relatively thin compared to the thickness of the substrate **230**. The substrate **230** may have a relatively thin thickness relative to a length and a width of the substrate **230**. In the present application, the width of a slide refers to the shorter distance across the front face of the slide, and the length of a slide refers to the longer distance across the front face of the slide. A width or narrow edge of a slide refers to an edge or side of the slide, not including the front face or the rear face, along the width direction. A length or long edge of a slide refers to an edge or side of the slide, not including the front face or the rear face, along the length direction. A top end of the slide refers to the end corresponding to the label **234**, and the bottom end refers to the end opposite the top end.

(106) FIGS. 3A-3J illustrate an example slide carrier **300** according to some embodiments of the

present disclosure. The slide carrier **300** may be utilized (or may be the same as), for example, as the slide carrier **208** illustrated in FIG. 2A. According to some embodiments, the slide carrier **300** may be utilized as a temporary or intermediate slide carrier, upon which a slide (e.g., the slide **204**) may be temporarily placed in a slide processing environment (e.g., the slide processing system **200**), for example, between operations of a slide manufacturing process.

(107) The slide carrier **300** may be formed of any suitable material that is solid enough to support a glass slide **204**, without substantial friction between the glass and the material of the slide carrier **300**. For example, in some embodiments, the slide carrier **300** may be formed of a plastic or polymer material, but embodiments according to the present disclosure are not limited thereto.

(108) According to some embodiments, as illustrated in FIG. 3A, the slide carrier **300** may be mounted on a surface or plane **302**. The surface **302** may be, for example, a work bench or table configured to hold one or more slide carriers **300** during the manufacturing process of one or more slides. In some embodiments, the surface **302** may be a planar surface parallel to a plane defined by an x-direction vector and a y-direction vector (i.e., an x-y plane). The plane defined by the x-direction vector and the y-direction vector may be perpendicular to the direction of gravity. In other words, a z-direction vector, which is normal or perpendicular to the surface **302** and/or the plane that is defined by the x-direction vector and the y-direction vector, may extend in a direction opposite to the force of gravity.

(109) As illustrated in FIG. 3A, according to some embodiments, the slide carrier may have one or more base supports **304**, which are configured to be mechanically connected and secured to the surface **302**. Although FIG. 3A illustrates two base supports **304a** and **304b**, embodiments according to the present disclosure are not limited thereto, and in some embodiments the slide carrier **300** may include a single base support **304**, or may include more than two base supports, without departing from the spirit and scope of embodiments according to the present disclosure. Additionally, in some embodiments the bottom of each base support **304** may be coplanar with the bottom of each other base support **304**, but embodiments according to the present disclosure are not limited thereto. For example, in some embodiments, the height (measured, for example, from the ground level, or the x-y plane) of the bottom of one base support **304** may be higher or lower than the height of another base support **304**.

(110) In addition to providing a mechanical structure for securing the slide carrier **300** to the surface **302**, the base supports **304** provide a platform or structure to support one or more pillars or vertical supports **306**, which provide vertical support in the z-direction for a slide platform **308**, configured to accommodate a slide (e.g., the slide **204**). According to some embodiments, the slide platform **308** may be a planar surface that is parallel to a plane that is rotated along both the x-direction and the y-direction relative to the x-y plane and/or the surface **302**.

(111) Further details of the relative orientation of the plane of the slide platform **308** are described with respect to FIGS. 3B and 3C. FIG. 3B shows a front view (when viewed in the y-direction of FIG. 3A) of the slide carrier **300** according to some embodiments. As illustrated in FIG. 3B, the plane of the slide platform **308** may be rotated about an axis that is parallel to the y-direction. For example, the plane of the slide platform **308** may be parallel to a plane (an X'-Y' plane) defined by a vector or direction X' and a vector or direction Y'. The X' direction may be at an angle relative to an axis that is parallel to the x-direction, such that the plane of the slide platform **308** is rotated at an angle θ , about an axis parallel to the y-direction, relative to the x-direction. According to some embodiments, the angle θ may be greater than 0 degrees and less than 90 degrees, but the angle θ may vary in different embodiments. According to some embodiments, the angle θ is greater than a first predetermined minimum angle and less than a first predetermined maximum angle. In some embodiments the first predetermined minimum angle may be an angle at which a value equal to or larger than the first predetermined minimum angle will cause a slide (e.g., the slide **204**) to move in the x-direction (and the X' direction) in view of the forces of gravity and friction, when the slide is placed on the slide platform **308**. The first predetermined maximum angle may be an angle at

which a value that is equal to or less than the predetermined maximum angle, a slide (e.g., the slide **204**) will not have stability issues, or risk falling off of the slide platform **308**.

(112) FIG. **3C** shows a right side view (when viewed in a direction opposite the x-direction of FIG. **3A**) of the slide carrier **300** according to some embodiments. As illustrated in FIG. **3C**, the plane X'-Y' plane of the slide platform **308** may be rotate about an axis that is parallel to the x-direction. For example, Y' direction may be at an angle relative to an axis that is parallel to the y-direction, such that the X'-Y' plane of the slide platform **308** is rotated at an angle α , about an axis that is parallel to the x-direction, relative to the y-direction. According to some embodiments, the angle α may be greater than 0 degrees and less than 90 degrees, but the angle α may vary in different embodiments. According to some embodiments, the angle α may be greater than a second predetermined minimum angle and less than a second predetermined maximum angle. In some embodiments the second predetermined minimum angle may be an angle at which a value equal to or larger than the second predetermined minimum angle will cause a slide (e.g., the slide **204**) to move in the direction opposite to the y-direction (and the direction opposite to the Y' direction) in view of the forces of gravity and friction, when the slide is placed on the slide platform **308**. The second predetermined maximum angle may be an angle at which a value that is equal to or less than the second predetermined maximum angle, a slide (e.g., the slide **204**) will not have stability issues, or risk falling off of the slide platform **308**.

(113) Referring to FIGS. **3B** and **3C**, the slide platform **308** may include elongated support portions **310** that extend or are elongated in the Y' direction, and may include one or more crossmember support portions **312** that extend between the support portions **310** in the X' direction. According to some embodiments, the top surface of each of the support portions **310** and the crossmember support portions **312** may be coplanar with each other and the X'-Y' plane. Collectively, the top surfaces of each of the support portions **310** and the crossmember support portions **312** may provide a platform or surface upon which a slide (e.g., the slide **204**) may be placed. The relative angles of the X'-Y' plane of the slide platform **308**, in addition to the force of gravity which will exceed the counterforces of friction according to some embodiments, may cause the slide (e.g., the slide **204**) to gravitate toward or register at the lowest corner or intersection point C. That is, because the X'-Y' plane of the slide platform **308** is rotated or tilted about the x-direction axis and the y-direction axis, when a slide is placed on the slide platform **308**, the slide may drift or migrate toward the lowest point on the slide platform **308** at the corner C.

(114) According to some embodiments, the width W1, which is the width measured from the outermost edges of the support portions **310** along the x-direction may be greater than the width of the slide (e.g., the slide **204**) to be placed on the slide platform **308**. Additionally, the width W2, which is the width measured from outside of the lower long edge of the slide platform **308** (e.g., the outside edge of the support portion **310b**) to the inside or closest edge of the opposite support portion **310** (e.g., the support portion **310a**), may also be greater than the width of the slide. Thus, because both the width W1 and the width W2 may be greater than the width of the slide, the risk of the slide falling or teetering off of the slide platform **308** may be relatively reduced or minimized. In other embodiments, however, the width W2 may be less than the width of a slide, and the crossmember support portion **312** may provide support to the slide **204** to prevent the slide from sliding or slipping between the space or opening between opposing support portions **310a** and **310b**.

(115) FIG. **3D** shows a left side view of the slide carrier **300** according to some embodiments. As shown in FIG. **3D**, when viewed from the left side of the slide carrier **300**, the upper surface of the support portion **310a** may be visible, but the support portion **310b** is below (e.g., at a lower height or elevation) compared to the support portion **310a**. Additionally, as illustrated in FIG. **3D**, according to some embodiments, there may be an opening or space between the pillars or vertical supports **306** in the y-direction. In some embodiments, there may also be an opening or space between vertical supports **306** in the x-direction. Thus, in some embodiments, the slide carrier **300**

may include two vertical supports **306** at or near opposing ends of the slide carrier **300**, or the slide carrier **300** may include additional vertical supports **306**, for example, four vertical supports **306** at or near different corners of the slide carrier **300**. By creating an opening between different vertical supports **306**, the slide carrier **300** may be able to utilize a reduced amount of material, with a relatively lower weight, while still maintaining structural integrity and strength.

(116) FIG. **3E** shows a left side perspective view of the slide carrier **300** according to some embodiments. As illustrated in FIG. **3E** (and also illustrated in FIG. **3B**) the slide carrier **300** may include a barrier or wall **314** that protrudes above the surface of the slide platform **308** at the lower long edge and at the lower short edge of the slide platform **308**. The barrier **314** may operate to prevent a slide from sliding off of the slide carrier **300** from the force of gravity. In some embodiments, one or more bumpers or protrusions **316** may extend or protrude from the barrier **314** (for example, with one or more spaces or gaps along the barrier **314** between the adjacent bumpers **316**), which may further assist with registering the slide in the intended location. In some embodiment the bumpers **316** may have a softer texture or may be more compressible than the material of the barrier **314**, which may prevent damage to the slide.

(117) Additionally, in some embodiments, the bumpers **316** may be elevated above the slide platform **308**, or may have a lip or bevel above the slide platform **308** to enable a slide to at least partially extend under a portion of the bumpers **316**, thereby providing additional force or pressure against the slide toward the slide platform **308** in order to prevent the slide from inadvertently slipping off of the slide platform **308**.

(118) Referring again to FIGS. **3E** and **3B**, in some embodiments, a connection structure **318** may extend between the opposing support portions **310a** and **310b** to provide a mechanical connection between the support portions **310a** and **310b**, as well as to improve the structural integrity of the overall slide carrier **300**. According to some embodiments, however, the upper surface of the connection structure **318** may be receded relative to the X'-Y' plane of the slide platform **308**. That is, in some embodiments, the connection structure **318** may be located below the surface of the slide platform **308**, which may reduce friction with the slide and reduce the likelihood of an edge of the slide unintentionally making contact with, or getting caught up by, the side wall of the connection structure **318** when the slide is being placed on the slide platform **308**.

(119) Additionally, in some embodiments, various locations on the slide platform **308** may be utilized as teaching positions for aligning an automated or robotic system. In some embodiments, additional, or alternative teaching positions may be located various known angles and locations relative to the slide platform **308** to enable the automated or robotic system to identify the location of the slide carrier **300** and the slide platform **308**.

(120) FIG. **3F** shows a rear perspective view of the slide carrier **300** according to some embodiments. As shown in FIG. **3F**, one or more of the base supports **304** may include an opening or hole **320**. The holes **320** may be utilized for securing the slide carrier **300** to the surface **302**, for example, by using a screw or bolt. Although FIG. **3F** shows two holes **320a** and **320b** located at each base support **304a** and **304b**, respectively, embodiments according to the present invention are not limited thereto, and in some embodiments, each base support **304** may include a plurality of openings or holes **320** for securing the slide carrier **300** to the surface **302**. Alternatively, in some embodiments, the openings or holes **320** may be omitted for one or more of the base supports **304**. Additionally, the slide carrier **300** may be mounted or secured to the surface **302** by utilizing any other suitable mechanical mounting mechanism. For example, in some embodiments, an adhesive, one or more clips, springs, grooves, teeth, etc., may be utilized to secure the slide carrier **300** to the surface **302**.

(121) Additionally, according to some embodiments as illustrated in FIG. **3F**, the support portions **310** may have narrow support portions **324a** and **324b**, at the higher elevation end of each support portion **310a** and **310b**, respectively. The top surfaces of the narrow support portions **324a** may be coplanar with the top surfaces of the support portions **310a** and **310b**, and the crossmember support

portion **312**. The distance between the narrow support portions **324a** and **324b**, however, may be less than the distance between the support portions **310a** and **310b**. According to some embodiments, the distance between the outside edges of each of the narrow support portions **324a** and **324b** may be less than the width of a slide when it is placed on the slide platform **308**.

(122) FIG. 3G shows a rear view of the slide carrier **300** according to some embodiments. As shown in FIG. 3G, in some embodiments, the outside width **W3** between the narrow support portions **324a** and **324b** may be less than the width of the slide carrier **300** (and the width between the support portions **310a** and **310b**). Additionally, as discussed above, in some embodiments, the width **W3** may be less than the narrow width of a slide when it is placed on the slide platform **308**. Thus, it may be relatively easy for a gripper of a mechanical arm to grip the slide in a pinching action P.sub.1 (described in more detail below) against the short or narrow width of the slide that is laying on the slide platform **308**.

(123) Additionally, as illustrated in FIG. 3G, there may be a space between the narrow support portions **324a** and **324b** that has a width **W4** that is less than the width **W3**. The space between the narrow support portions **324a** and **324b** may enable the gripper of a mechanical arm to grip the flat sides of the slide in a pinching action P.sub.2 when the slide is resting on the slide platform **308**. For example, the space between the narrow support portions **324a** and **324b** may accommodate a portion of the gripper to fit below the slide, and another portion of the gripper may be maneuvered above the slide, such that the pinching action P.sub.2 causes the gripper to put pressure on opposing flat surfaces of the slide in order to pick up or maneuver the slide.

(124) FIG. 3H shows a top view of the slide carrier **300** according to some embodiments. FIG. 3I shows a bottom view of the slide carrier **300** according to some embodiments. As shown in FIG. 3H, in addition to access to pinching or picking up the slide from the perpendicular pinching actions P.sub.1 and P.sub.2 described with respect to FIG. 3G, in some embodiments, there may be an opening **326** at the opposite end of the slide carrier **300** relative to the narrow support portions **324a** and **324b**. The opening **326** may be configured to accommodate a gripper of a mechanical arm in order to pick up a slide resting on the slide platform **308** in a pinching action that is perpendicular to the X'-Y' plane.

(125) FIG. 3J is a flow diagram illustrating a method of manufacturing a pathology slide in a slide processing system, according to some embodiments. Although various operations are illustrated in FIG. 3J as part of the method of manufacturing a pathology slide in a slide processing system, embodiments according to the present disclosure are not limited to the operations illustrated in FIG. 3J. For example, some embodiments may include additional operations or fewer operations without departing from the spirit and scope of embodiments according to the present disclosure.

(126) As described above, the manufacturing process starts, and then, at operation **330**, a gripper may grip a slide in a first orientation (e.g., any of the orientations described above with respect to FIGS. 3A-3I). At operation **332**, the gripper may place the slide on a slide platform of a slide carrier (e.g., the slide carrier **300**). At operation **334**, the slide may be allowed to shift or move (e.g., under the force of gravity) locations on the slide platform to register the slide at a predetermined location on the slide platform. At operation **336**, the slide may be gripped with the gripper in a second orientation (e.g., any of the operations described above with respect to FIGS. 3A-3I), and the second orientation may correspond, for example, to a component (e.g., a slide carrier or slide processing station). At operation **338**, the gripper may place the slide in the component of the slide processing system using the gripper with the slide in the second orientation to enable a slide processing operation that corresponds to the component to be performed on the slide by the slide processing system.

(127) Thus, as described above, embodiments according to the present disclosure may include a slide carrier that enables a slide to be picked up or placed down during intermediate manufacturing operations from various angles or orientations. When the slide is placed on the slide carrier, the slide will gravitate to a known, fixed position and orientation, enabling the gripper and mechanical

arm to pick up the slide again relatively easily. The slide carrier utilizes compound angles to bias the edges of the slide toward a known location (e.g., the bottom right edge) of the slide carrier. The resting surface of the slide carrier may be minimized or reduced in order to reduce friction on the faces of the slide, and angles of the slide carrier may be optimized or designed to increase potential energy and help to ensure the slide will move under gravity. Additionally, various features are incorporated into the slide carrier to enable various gripper and suction cup designs for picking up and dropping off slides. Additionally, registration features may be added to facilitate teaching positions for aligning to an automated or robotic system. According to some embodiments, an edge detection system (e.g., by utilizing a camera or sensor on a mechanical arm or gripper) may be utilized to detect the location of the slide.

(128) FIGS. 4A-4M illustrate a mechanical arm gripper according to some embodiments of the present disclosure. As described above with respect to FIGS. 2 and 3, according to some embodiments of the present disclosure, a mechanical arm with a gripper connected to an end of the mechanical arm may be configured to pick up and manipulate a slide in a slide processing or manufacturing environment.

(129) FIG. 4A illustrates an example of a gripper **400** according to some embodiments. The gripper **400** may be the same as or similar to (or one example of) the gripper **216** illustrated in the slide processing or manufacturing system **200** of FIG. 2A.

(130) The gripper **400** according to some embodiments may be configured to grip or manipulate a slide in various positions or orientations, in order to enable the slide to interface with various different carriers and machines in the process of manufacturing a pathology slide. Thus, embodiments according to the present disclosure may reduce the need to change grippers or tools, and/or utilize multiple mechanical arms during the manufacturing process.

(131) The gripper **400** may be formed of any suitable material that is solid enough to support a glass slide **204**, without a substantial risk of the glass slide or the gripper **400** being damaged. For example, in some embodiments, the gripper **400** may be formed of a plastic or polymer material, but embodiments according to the present disclosure are not limited thereto.

(132) As shown in FIG. 4A, the gripper **400** may include two or more gripper fingers **402**. Although FIGS. 4A-4M are described in the context of two gripper fingers **402a** and **402b**, embodiments according to the present disclosure are not limited thereto, and some embodiments may include additional gripper fingers **402** without departing from the spirit and scope of embodiments according to the present disclosure. In the present Specification, in some instances, the reference numeral **402** may be used to refer generally to each of the gripper fingers **402a** and **402b** (and any additional gripper fingers that may be utilized in some embodiments).

(133) Additionally, according to some embodiments, the gripper **400** (or the mechanical arm to which the gripper **400** is attached) may include additional components configured to be utilized in connection with the manipulation and maneuvering of a slide during a slide manufacturing process. For example, in some embodiments, the gripper **400** (or the mechanical arm to which the gripper **400** is attached) may include a suction cup for creating a suction force against a slide. Additionally, in some embodiments, the gripper **400** (or the mechanical arm to which the gripper **400** is attached) may include a camera or sensor (e.g., an optical sensor, a pressure sensor, etc.) configured to identify or measure the location of a slide, a slide carrier, a surface, or the like.

(134) As shown in FIGS. 4A-4M, the gripper **400**, using the gripper fingers **402a** and **402b**, may be configured to apply a pinching or inward force **P3** toward a center line **CENTER**, which bisects the gripper **400**. The gripper fingers **402a** and **402b** may be positioned an equal distance from the center line **CENTER**, and may be configured to move toward and away from the center line **CENTER** in coordination with each other (e.g., at a same speed and with a same force).

(135) Referring to FIGS. 4A-4M, the gripper fingers **402a** and **402b** may be mounted to an actuator **404** configured to move the gripper fingers **402a** and **402b** inward (toward the center line **CENTER**) and outward (away from the center line **CENTER**) in order to open or close a space

between the gripper fingers **402a** and **402b**. The actuator **404** may be connected to a mechanical arm (e.g., the mechanical arm **202** shown in FIG. 2A), such that collectively, the mechanical arm **202**, the actuator **404**, and the gripper **400** may operate to manipulate a slide or move the slide from one location to another location. Each gripper finger **402a** and **402b** may have the same or similar shape. For example, in some embodiments, the gripper finger **402b** may be the same as the gripper finger **402a**, except rotated 180 degrees relative to the gripper finger **402a**, although embodiments according to the present disclosure are not limited thereto.

(136) FIG. 4A illustrates the gripper **400** holding a slide in a first gripping mode or configuration, in which the gripper fingers **402a** and **402b** operate to provide a pinching force **P3** widthwise (along an axis parallel to a width direction of the slide **204**), against the long edges of the slide **204**.

(137) In the first gripping mode, the gripper **400** may be configured to pick up and/or maneuver the slide **204** in an orientation in which the center line between the gripper fingers **402a** and **402b** extends along the front face of the slide **204** in parallel with the long edges of the slide **204**, with the center line bisecting the short edges of the slide **204**.

(138) The gripper fingers **402a** and **402b** may have a first portion **406** that extends generally in a direction away from the actuator **404**. For example, in some embodiments, the first portion **406** may extend in a direction parallel (or generally parallel) to the center line CENTER. The first portion **406** of each gripper finger **402a** and **402b** may have an interior edge **408a** and **408b**, respectively, where the interior edges **408a** and **408b** are interior with respect to the opening or space between the gripper fingers **402a** and **402b**.

(139) In some embodiments, the interior edges **408a** and **408b** may be elongated in a direction parallel to the center line CENTER. In some embodiments, the interior edges **408a** and **408b** may extend parallel to each other, and may each have a groove therein, which are each adapted to receive the top (or bottom) of the long edges of a slide as shown in the first gripping mode shown in FIG. 4A. The groove in the interior edges **408a** and **408b** is illustrated and described in more detail below.

(140) When the pinching or inward force **P3** is applied to the long edges of the slide, the gripper **400** may be able to pick up the slide **204** and move the slide around in 3-dimensional space.

(141) FIG. 4B illustrates the gripper **400** holding a slide in a second gripping mode or configuration, in which the gripper fingers **402a** and **402b** operate to provide a pinching force or pressure **P4** lengthwise (along an axis parallel to a length direction of the slide **204**) against the short or width edges of the slide **204**. That is, the pressure or force **P4** may be in a direction toward (e.g., perpendicular to) the center line CENTER. In the second gripping mode, the gripper **400** may pick up and/or maneuver the slide **204** in an orientation in which the center line CENTER between the gripper fingers **402a** and **402b** may extend through the front and rear faces of the slide **204**, halfway between the short edges of the slide **204**. That is, a direction that is perpendicular or normal with respect to the planes of the front and rear faces of the slide **204** may be parallel to the center line CENTER, such that the front or rear face of the slide **204** faces toward the actuator **404**.

(142) The gripper fingers **402a** and **402b** may have a second portion **410** that extends from the first portion **406** at an angle that is greater than zero degrees and less than 90 degrees relative to the extension direction of the first portion **406**. For example, if the first portion **406** extends in a direction **D1** away from the actuator **404**, the second portion **410** may extend in a direction **D2** away from the first portion **406**, where an angle γ between the direction **D1** and the direction **D2** is less than 90 degrees and greater than zero degrees. Accordingly, an end **412**, which is the furthest point of the second portion **410** from the actuator **404**, may be further away from the actuator **404**, and may also be further away from the center line CENTER, compared to the interior edges **408a** and **408b** of the first portion **406**.

(143) The end **412** of the second portion may include a protrusion or tooth **414** that extends toward the center line CENTER, such that a groove or cavity **416** is created between the protrusion **414** and the main body **418** of the second portion **410**. Further details of the groove or cavity **416** and

the protrusion **414** are illustrated and described in more detail below.

(144) Because the ends of the second portion **410** are both further away from the actuator **404**, and further away from each other, compared to the interior edges **408a** and **408b**, a wider space may be created between the ends of the second portion **410**, relative to the space between the interior edges **408a** and **408b**. Additionally, the groove **416** between the protrusion **414** and the main body **418** of the second portion **410** may accommodate the width or short edges of the slide **204** such that the gripper **400** can pick up and/or maneuver the slide **204** with the front or rear face of the slide **204** facing toward the actuator **404**.

(145) FIG. 4C illustrates the gripper **400** holding a slide in a third gripping mode or configuration, according to some embodiments. In the third gripping mode, the gripper fingers **402a** and **402b** operate to provide a pinching force or pressure **P5** lengthwise (along an axis parallel to a length direction of the slide **204**) against the width or short edges of the slide. That is, the pressure or force **P5** may be in a direction toward (e.g., perpendicular to) the center line CENTER. In the third gripping mode, the gripper **400** may pick up and/or maneuver the slide **204** in an orientation in which the center line CENTER between the gripper fingers **402a** and **402b** may extend through the long edges of the slide **204**, halfway between the short edges of the slide **204**. That is, in contrast to the second gripping mode, in which either the front or rear surface of the slide **204** faces toward the actuator **404**, in the third gripping mode, one of the long edges of the slide **204** may face the actuator **404**, with the center line CENTER bisecting the front and rear faces, and the front and rear faces facing a direction perpendicular to the extension line of the center line CENTER.

(146) According to some embodiments, the second portion **410** of each gripper finger **402a** and **402b** may include interior edges **420a** and **420b**, respectively, that extend in parallel to the extension directions of the interior edges **408a** and **408b**. The interior edges **420a** and **420b** may extend in parallel to each other and to the interior edges **408a** and **408b**. The interior edges **420a** and **420b** may be interior with respect to the opening or space between the gripper fingers **402a** and **402b**, and may be elongated or extend in a direction parallel to the center line CENTER. Each interior edge **420a** and **420b** may face the center line, and may include a groove, which is adapted or configured to accommodate or receive the top or bottom portion of the width edges of the slide **204**. The grooves of the interior edges **420a** and **420b** are illustrated and described in more detail below.

(147) FIG. 4D illustrates the gripper **400** holding a slide in a fourth gripping mode or configuration, according to some embodiments. In the fourth gripping mode, the gripper fingers **402a** and **402b** operate to provide a pinching force or pressure **P6** lengthwise (along an axis parallel to a length direction of the slide **204**) against the short or width edges of the slide **204**.

(148) That is, the pressure or force **P6** may be in a direction toward (e.g., perpendicular to) the center line CENTER. In the fourth gripping mode, the gripper **400** may pick up and/or maneuver the slide **204** in an orientation in which the center line CENTER between the gripper fingers **402a** and **402b** may extend through the front and rear faces of the slide **204**, halfway between the short edges of the slide **204**. That is, a direction that is perpendicular or normal with respect to the planes of the front and rear faces of the slide **204** may be parallel to the center line CENTER, such that the front or rear face of the slide **204** faces toward the actuator **404**.

(149) The gripper fingers **402a** and **402b** may have a protrusion **426** that extends from the end **412** in a direction parallel to the center line CENTER, below the protrusion **414**. In some embodiments, the protrusion **414** may be configured to contact the front or rear face of the slide **204** in the fourth gripping mode, and the protrusions **426** may be configured to apply the pressure or force **P6** toward the center of the slide **204**.

(150) The fourth gripping mode may be configured to be utilized in an operation in which the slide **204** is manipulated with respect to a slide carrier **430** (the slide carrier **430** may be the same as, or similar to, any of the slide carriers **208**, **214a**, or **214b** described with respect to FIG. 2A). For example, the slide carrier **430** may have one or more cavities **432** that are configured to

accommodate the slide **204**. For example, the cavities **432** may have a width, length, and depth that correspond to (e.g., are slightly larger within a predetermined or acceptable tolerance than) the width, length, and thickness of the slide **204**.

(151) Because the protrusion **426** may extend past the length of the slide **204**, however, in order to enable the gripper **400** to safely place the slide **204** into the cavity **432** and/or access a firm/sufficient grip on the slide **204** when the slide **204** is already in the cavity **432**, the cavity **432** may include a first cavity extension **436** and a second cavity extension **438**. The first and second cavity extensions **436** and **438** may be located at opposite ends of the cavity **432** in the length direction. The cavity extensions **436** and **438** may have a size or footprint that is larger than that of the protrusion **426**, such that the protrusion **426** can fit within the cavity extensions **436** and **438** when the gripper **400** picks up and/or sets down the slide **204** within the cavity **432**. In some embodiments, the cavity extensions **436** and **438** may have a circular or rounded edge in a plan view, but embodiments according to the present disclosure are not limited thereto. For example, the cavity extensions **436** and **438** may have a rectangular shape, or a shape that follows the contour of the gripper **400** in a plan view, or any other suitable shape capable of accommodating the protrusion **426**.

(152) FIG. **4E** illustrates the gripper **400** manipulating a slide using an exterior groove or cavity, according to some embodiments. According to some embodiments, the gripper fingers **402** may have an exterior groove **440** extending in a direction perpendicular to the center line CENTER. The exterior groove **440** may be located, for example, on the opposite side of the gripper fingers **402** relative to the space between the gripper fingers **402**. The exterior groove **440** may also be located on an opposite side of the second portion **410** relative to the groove **416**.

(153) The exterior groove **440** may have a shape configured to receive or accommodate one of the edges of the slide **204**, in order to push or provide a force against the slide **204** in a horizontal direction H and/or a vertical direction V relative to a plane in parallel to a front or rear face of the slide **204**. For example, as illustrated in FIG. **4E**, a slide carrier **442** may be utilized to hold or secure the slide **204**. The slide carrier **442** may be, for example, the same as the slide carriers **208**, **214a**, or **214b**, described with respect to FIG. **2A**. The slide carrier **442** may include a plurality of cavities **444** configured or shaped to accommodate the slide **204**. Each cavity **444** may additionally include a cavity extension **446** that is adjacent to the width edge of the cavity **444**, and which is configured or shaped to accommodate the end of the gripper finger **402** and to enable the groove **440** to engage with the width edge of the slide **204**.

(154) In some embodiments, the slide carrier **442** may include a spring or clip requiring a force or pressure in the horizontal direction H in order to remove the slide **204** from the slide carrier **442** or to insert the slide **204** into the slide carrier **442**. Thus, the exterior groove **440** may be configured to engage the edge of the slide **204**, such that the edge of the slide **204** fits within the curvature of the groove **440**. The actuator **404** and the gripper finger **402** may then move in the horizontal direction H and/or the vertical direction V to cause the slide **204** to engage or disengage with the slide carrier **442**.

(155) FIG. **4F** is a top front perspective view of a single gripper finger **402**, according to some embodiments. FIG. **4G** is a top rear perspective view of a single gripper finger **402** according to some embodiments. As shown in FIGS. **4F** and **4G**, the gripper finger **402** may include a mounting plate **450**. The mounting plate **450** may include a plurality of holes **452** formed therethrough. Thus, the gripper finger **402** may be mounted or mechanically connected to the actuator **404** by the mounting plate **450**, with one or more screws, bolts, etc., passing through the holes **452** to fasten the gripper finger **402** to the actuator **404**. Additionally, some embodiments may utilize any other suitable fastening mechanism to mechanically connect and secure the gripper finger **402** to the actuator **404**.

(156) As illustrated in FIGS. **4F** and **4G**, and as described above, each of the grooves **416** and **440** may have a shape that is configured to accommodate the edges (e.g., the width edge) of a slide **204**.

For example, the width or distance across the grooves **416** and **440** may be greater than the thickness (measured from the front face to the rear face) of the slide. In some embodiments, at least one of the groove **416** or the groove **440** may have a curvature (for example, that conforms to, or can accommodate, the edges of the slide **204**). Additionally, in some embodiments, at least one of the groove **416** or the groove **440** may have a rectangular shape and/or sloped sidewalls. The grooves **416** and **440** are configured to engage the slide **204** in a configuration in which either the front or rear face of the slide **204** is oriented toward the actuator **404** (and the mounting plate **450**). (157) Similarly, as illustrated in FIGS. **4F** and **4G**, and as described above, each of the interior edges **408** and **420** may have a shape that is configured to accommodate the edges of a slide **204**. In particular, the interior edge **408** may be utilized to accommodate the top or bottom of the length edge of the slide **204** (as shown in FIG. **4A**), and the interior edge **420** may be utilized to accommodate the width edge of the slide **204** (as shown in FIG. **4C**). For example, the distance across the groove or cavity of the interior edge **408** and the interior edge **420** may be greater than the thickness of the slide. The interior edges **408** and **420** are therefore configured to engage the slide **204** in a configuration in which either the width edge or the length edge of the slide **204** is oriented toward the actuator **404** (and the mounting plate **450**).

(158) FIG. **4H** is a bottom view of a single gripper finger **402** according to some embodiments. FIG. **4I** is a close-up view of the section A of FIG. **4H** according to some embodiments. FIG. **4J** is a close-up view of the section B of FIG. **4H** according to some embodiments. Thus, FIGS. **4I** and **4J** illustrate further details of the interior edges **408** and **420**, according to some embodiments.

(159) Referring to FIGS. **4H** and **4I**, and as discussed above, the interior edge **408** may be configured to accommodate the edge of a slide **204**. Thus, the interior edge **408** may include a groove or cavity **460**. The cavity **460** may include a first portion **462** that has a width or distance **D1** across the cavity **460** at the first portion **462** that is greater than the thickness of the slide **204**. Thus, the first portion **462** of the cavity **460** may be configured to accommodate the thickness of the slide **204** to enable the gripper **400** to put the pressure or force against the edge of the slide **204**. The cavity **460** may further include a second portion **464** that has a width or distance **D2** at an opening of the cavity **460** that is wider than the width **D1**, and then tapers down to the width **D1** at the junction between the first portion **462** and the second portion **464**. Thus, because the opening of the cavity **460** is wider than the width **D1**, and then tapers down to the width **D1**, the cavity **460** in some embodiments may enable the slide **204** to be inserted into the cavity **460** with a higher degree of tolerance.

(160) Similarly, referring to FIGS. **4H** and **4J**, and as discussed above, the interior edge **420** may be configured to accommodate the edge of a slide **204**. Thus, the interior edge **420** may include a groove or cavity **470**. The cavity **470** may include a first portion **472** that has a width or distance **D3** across the cavity **470** at the first portion **472** that is greater than the thickness of the slide **204**. Thus, the first portion **472** of the cavity **470** may be configured to accommodate the thickness of the slide **204** to enable the gripper **400** to put the pressure or force against the edge of the slide **204**. The cavity **460** may further include a second portion **474** that has a width or distance **D4** at an opening of the cavity **470** that is wider than the width **D3**, and then tapers down to the width **D3** at the junction between the first portion **472** and the second portion **474**. Thus, because the opening of the cavity **470** is wider than the width **D3**, and then tapers down to the width **D3**, the cavity **470** in some embodiments may enable the slide **204** to be inserted into the cavity **470** with a higher degree of tolerance.

(161) FIG. **4K** is a side view of a single gripper finger **402** according to some embodiments. FIG. **4L** is a close up view of the section C of FIG. **4K** according to some embodiments. Thus, FIG. **4L** shows further details of the protrusions **414** and **426**, and the grooves **416** and **440** according to some embodiments.

(162) As shown in FIGS. **4K** and **4L**, the protrusion **414** may extend in a direction toward the center line CENTER shown, for example, in FIG. **4A**. The protrusion **414** may operate as a support

to the slide **204**, when the edge of the slide **204** is inserted or accommodated within the groove **416**. Additionally, as described with respect to the interior edges **408** and **420**, the groove **416** may have a width at a deeper portion that corresponds to, or is greater than, the thickness of the slide **204**, and the groove **416** may have an even wider width at the opening (formed or defined by the main body of the gripper finger **402** and the protrusion **414**) that is wider and then tapers down to a smaller width.

(163) Additionally, as shown in FIG. **4L**, the protrusion **426** may extend vertically below the protrusion **414**, and may have an edge **476** that extends in parallel to the center line CENTER (shown, for example, in FIG. **4A**). The edge **476** may provide a surface to provide a compressive or lateral force against a slide **204**.

(164) The exterior groove **440** may be positioned above the protrusion **426**. An edge or shelf **478** may operate as a support or platform for the front or rear face of the slide **204**. The other surfaces of the groove **440** may provide a surface for pushing against the edge of the slide **204**.

(165) FIG. **4M** is a flow diagram illustrating a method of manufacturing a slide in a slide processing system, according to some embodiments. Although various operations are illustrated in FIG. **4M** as part of the method of manufacturing a pathology slide in a slide processing system, embodiments according to the present disclosure are not limited to the operations illustrated in FIG. **4M**. For example, some embodiments may include additional operations or fewer operations without departing from the spirit and scope of embodiments according to the present disclosure.

(166) As described above, the manufacturing process starts, and then, at operation **490**, a gripper may grip a slide in a first orientation (e.g., any of the orientations described above with respect to FIGS. **3A-4L**) using a first groove of a first gripper finger and a first groove of a second gripper finger. At operation **492**, the gripper may place the slide on a slide in a first component of the slide processing system in the first orientation. At operation **494**, the gripper may grip the slide in a second orientation using a second groove of the first gripper finger and a second groove of the second gripper finger. At operation **496**, the slide may be placed in a second component of the slide processing system in the second orientation.

(167) Therefore, according to some embodiments, the gripper may be utilized in conjunction with a slide carrier described above or the slide carrier cassette described below to transfer a slide from the slide carrier and/or the slide carrier cassette to another component or apparatus within the slide processing system **200**.

(168) Thus, as described above, embodiments according to the present disclosure may include a gripper capable of being utilized with multiple different carriers or slide manufacturing machines, each of which may have different requirements as far as orientation and gripping position when slides are inserted or removed, as part of a pathology slide manufacturing process or workflow. That is, automation of transferring slides between carriers and to positions within instruments requires interfacing with slides and carriers or machines in many different ways. Robotic arms and actuators may otherwise be limited in gripper tool attachments, making it difficult to perform different transfer tasks with a single gripper tool. Thus, embodiments according to the present disclosure may reduce tool changeover or multiple mechanical arm requirements.

(169) According to various other embodiments, a slide carrier cassette may be utilized to enable or facilitate processing of multiple slides in coordination and/or concurrently. The slide carrier cassette may also have features for enabling self-alignment of slides, similar to the features described above with respect to FIGS. **3A-3J**, and as described below. FIG. **5A** is a front perspective view of a slide carrier cassette according to some embodiments. FIG. **5B** is a rear perspective view of a slide carrier cassette according to some embodiments. A slide carrier cassette **500** according to various embodiments has a main body or housing **501** having a plurality of slots **502-1** through **502-n** formed therein, with openings of the slots **502-1** through **502-n** being located at a top side of the main body **501**, where “n” is a natural number greater than 1.

(170) The slide carrier cassette **500** may be formed of any suitable material that is solid enough to

support a glass slide **204**, without substantial friction between the glass and the material of the slide carrier cassette **500**. For example, in some embodiments, the slide carrier cassette **500** may be formed of a plastic or polymer material, but embodiments according to the present disclosure are not limited thereto.

(171) The slide carrier cassette **500** may include any suitable number of slots **502-1** through **502-n**. The slots **502-1** through **502-n** may be arranged or aligned adjacent to each other along an alignment direction A, and each slot **502-1** through **502-n** may be configured to receive or accommodate a single slide **204**. The front and rear faces of each slide **204**, when inside one of the slots **502-1** through **502-n**, is oriented toward the front or rear surface of the other slides **204** in other slots. That is, according to some embodiments, planes of the front and rear faces of each slide **204** that is inserted into the slots **502-1** through **502-n** is parallel the planes of the front and rear faces of each other slide **204**. Additionally, the bottom of each slot **502-1** through **502-n** may be aligned along the alignment direction A, such that each slot **502-1** through **502-n** may be a same distance or height from the bottom edge **508** of the slide carrier cassette **500**.

(172) Each slot **502-1** through **502-n** may be formed at a slanted angle (e.g., an angle greater than zero degrees and less than 90 degrees) relative to a plane that is perpendicular to the direction of gravity. Additionally, in some embodiments, each slot may be rotated (e.g., when viewed in a front view or a rear view) to either the left or right, such that a slide **204**, when placed in one of the slots **502-1** through **502-n** will register or self-align to a known reference location. The orientation and angle of the slots **502-1** through **502-n** is illustrated and described in more detail below.

(173) The slide carrier cassette **500** may further include one or more protrusions **504** (e.g., at a top surface of the slide carrier cassette **500**). The slide carrier cassette **500** may have a generally rectangular shape in a plan view (e.g., when viewed from a direction above the slide carrier cassette **500** or a direction parallel to gravity), and the protrusions **504** may be located around the edges of the slide carrier cassette **500** (e.g., at or near each corner). The protrusions **504** may be utilized to provide a fixed or semi-permanent mounting mechanism to a slide carrier cassette queue or storage rack (shown in more detail below), and may include, for example, a hole for utilizing a screw or bolt (or other suitable mechanical fastening mechanism) to secure the slide carrier cassette **500** to the queue or storage rack.

(174) In some embodiments, the slide carrier cassette **500** may include a spring (e.g., a living hinge spring) or clip **510**. The spring **510** may provide a spring force in an external direction away from the body of the slide carrier cassette **500**, such that when the slide carrier cassette **500** is inserted into a queue or storage rack (described in more detail below), the spring **510** may create a retention force to enable the slide carrier cassette **500** to be retained or held by queue or storage rack. The spring **510** may further enable alignment and proper positioning and registration of the slide carrier cassette **500** within the queue or storage rack.

(175) The slide carrier cassette **500** may further include a tab handle **512** at a top surface (or top side) **514** of the slide carrier cassette **500**. The tab handle **512** may extend above the top surface **514** of the slide carrier cassette **500**, and may have a generally planar shape (although embodiments are not limited thereto). The tab handle **512** may operate as a handle or grasping structure for a human or mechanical gripper to grip onto the structure of the slide carrier cassette **500**, for example, when maneuvering/transporting the slide carrier cassette **500**, or when inserting the slide carrier cassette **500** into a queue or storage rack, or removing the slide carrier cassette **500** from a queue or storage rack.

(176) When the tab handle **512** has a generally planar shape, the tab handle **512** may have a front face FF and a rear face RF. The front face FF may face toward a front of the slide carrier cassette **500**, and may have space or room to accommodate, for example, a label or unique identifying information for the slide carrier cassette **500**. The rear face RF may face toward the slots **502-1** through **502-n** and the rear (or back) side of the slide carrier cassette **500**.

(177) FIG. 5C is a front view of a slide carrier cassette according to some embodiments.

(178) As shown in FIG. 5C, and as described above, the front face FF of the tab handle 512 may include a label area 516, configured to accommodate a label or unique identifying information corresponding to the slide carrier cassette 500 and/or the slides held by the slide carrier cassette 500. For example, the label may include a barcode, QR code, or alphanumeric characters to provide identifying information regarding the nature and content of the slide carrier cassette 500.

(179) At the bottom of the slide carrier cassette 500, below the bottom edge 508, the slide carrier cassette 500 may include a plurality of feet or posts 520, that may be utilized to provide stability when the slide carrier cassette 500 is resting on a planar surface or a substrate with the bottom edge 508 facing toward the substrate. Additionally, the posts 520 may be utilized to align the slide carrier cassette 500 into a fixture such as a queue or storage rack.

(180) FIG. 5D is a cross-sectional view taken along the line A-A of FIG. 5C. FIG. 5E is a close-up view of the section B of FIG. 5D according to some embodiments. Referring to FIGS. 5D and 5E, and as described above, the slide carrier cassette 500 may include one or more springs or clips 510 located at one or more locations around the sides of the slide carrier cassette 500. In some embodiments, the spring 510 may be a living hinge spring, in which a narrow piece is connected to the body of the slide carrier cassette 500 at one end, and a ball or protrusion extends from the spring 510 at the opposite end. The protrusion of the spring 510 may fit or be accommodated within a corresponding hole or cavity on a storage rack or queue to enable a “snap-in” retention of the slide carrier cassette 500, and also to enable the slide carrier cassette 500 to be aligned in the proper position and orientation.

(181) Additionally, as described above, the plurality of slots 502-1 through 502-n may be aligned with each other in the slide carrier cassette 500 along an alignment direction A such that, when slides are inserted into the slots 502-1 through 502-n, the faces of the slides face toward the adjacent slide, and the edges of the slides face toward the sides of the slide carrier cassette 500. Each slot 502-1 through 502-n may have a thickness T1 that corresponds to (e.g., is slightly greater than) the thickness of the slide, such that movement of a slide that is inserted into the slide carrier cassette 500 in the alignment direction (or opposite to the alignment direction) A may be minimized or relatively reduced. Additionally, each slot 502-1 through 502-n may have a larger thickness T2 at the opening to enable relatively easier insertion of a slide into the slot and to accommodate larger tolerance variations. Each slot 502-1 through 502-n may have a chamfer or tapering portion at the junction between the smaller thickness T1 and the larger thickness T2.

(182) According to some embodiments, each slot may be angled at an angle σ relative to the alignment direction A, where σ is less than 90 degrees, and greater than 0 degrees. In embodiments in which σ is less than 90 degrees, it may be possible to more-easily read the label on each slide that is inserted into the slide carrier cassette 500.

(183) According to some embodiments, each slot 502-1 through 502-n may be separated by a set distance, the size of which may be a function of the size of a gripper to be utilized to insert and retrieve slides from the slide carrier cassette 500.

(184) FIG. 5F shows an example of a queue or storage rack for a plurality of slide carrier cassettes according to some embodiments. For example, according to some embodiments, a slide carrier queue 530 may include a plurality of cavities 532 arranged, for example, in a plurality of rows and columns. Although FIG. 5F illustrates a queue 530 having four rows and five columns of cavities 532, embodiments according to the present disclosure are not limited thereto, and a queue 530 according to the present disclosure may have any suitable number of rows and columns of cavities 532. Additionally, embodiments according to the present disclosure are not limited to the cavities 532 being arranged in columns and rows, various embodiments may include various different arrangements of cavities according to the design of the queue 530.

(185) According to some embodiments, each cavity 532 may be configured to accommodate the shape of the slide carrier cassette 500. Thus, in some embodiments the cavity 532 may have interior edges that conform to or correspond to the shape of the slide carrier cassette 500.

(186) According to some embodiments, the cavities **532** may be formed (e.g., machined) at an angle (e.g., rotated clockwise or counterclockwise to an angle greater than zero degrees, and less than 90 degrees) relative to a vertical line **V** (extending, for example, in parallel to the direction of gravity, or in the column direction), as illustrated in FIG. 5F. Because the slide carrier cassette **500** is mounted in the queue **530** at angle relative to the vertical line **V**, and because the slots **502-1** through **502-n** (described and labeled, for example, in FIG. 5A) are also slanted at an angle relative to the bottom edge **508** of the slide carrier cassette **500**, the slides may automatically (i.e., due to the force of gravity) bias to a known location. For example, various slides manufactured by different manufacturers or with slightly different formats, may have variations in width, thickness, length, and features (e.g., edges, corners, bevels, etc.) Such variations may be present in the same slide processing environment, due to receiving slides from different sources (e.g., different medical practices from all around the world).

(187) This known bias location allows for slides to be re-gripped at a predetermined or known reference point or location. In other words, the effect of variations of slide sizes and locations, when inserted into the slots **502-1** through **502-n**, may be reduced because the resting location of the slide will be at a known location.

(188) The queue **530** may operate as either an intermediate or long term storage location for slides. For example, as slides are inserted into individual slots of individual slide carrier cassettes **500**, other spaces may remain open, leaving space for other slides that are manufactured later in time to be included in the queue **530**. Once all slides in a batch have been manufactured, the slides can be retrieved from the queue **530** and positioned in a folio or case for longer term storage or to be transported (e.g., for review by a pathologist). The location of slides in individual slots may be managed according to a slot allocation scheme that is organized and controlled by a computer, and also slides may be retrieved manually, for example, by using the labels on individual slides and slide carrier cassettes, as discussed above.

(189) FIG. 5G shows an example of a queue or storage rack with a cylindrical shape according to some embodiments. As shown in FIG. 5G, according to some embodiments, instead of being arranged in a planar or linear matrix as shown in FIG. 5F, the queue **530** may be arranged in a circular or cylindrical configuration. In the cylindrical configuration shown in FIG. 5G, each row of cavities **532** may be formed in a circular shape around a central axis. In some embodiments, the queue **530** may be configured to be rotated around the central axis (e.g., using a rotary system on which the queue **530** may be mounted). The cylindrical configuration may reduce the amount of space needed to access the queue **530** by a mechanical arm, and may enable the queue **530** to be rotated to present individual slides or queues **530** to the mechanical arm, which may increase processing time.

(190) FIG. 5H shows an example of a queue or storage rack in a stacked configuration according to some embodiments. As shown in FIG. 5H, the queue **530** plurality of layers of cylindrical storage queues may be arranged in a stacked or stackable configuration according to some embodiments. Each layer or level of the queue may include a plurality of cavities **532** or openings that are arranged in a cylindrical or circular shape around a central axis, and which are configured to receive a slide carrier cassette **500**.

(191) Individual slide carrier cassettes **500** can be removed or inserted as desired, and individual layers of the queue **530** may also be removed or added as desired.

(192) The ability to remove individual slide carrier cassettes **500** from the queue **530** as shown in FIGS. 5F-5H may enable manual intervention or interaction with groups of slides, as well as enable cleaning and replacement of cassettes for maintenance over time. Using slide carrier cassettes **500** in the stackable configuration of the queue **530**, as shown in FIG. 5H may enable relatively simpler machined features, and may enable the slide carrier cassettes to be molded or 3-dimensional printed, which may enable the slide carrier cassettes to be manufactures at a relatively lower cost. Additionally, because the slide carrier cassettes **500** are removable from the queue **530**, they may

be utilized in other processes or instruments, and additionally the individual slides may be transported to different environments without removing them from the slide carrier cassette.

(193) FIG. 5I shows an example of a slide processing system **540**, according to some embodiments. The slide processing system **540** may be the same as, or similar to, the slide processing system described above with respect to FIGS. 2-5 (e.g., the slide processing system **200**), and may be utilized as part of the pathology system **100** (e.g., at operations **108** and **110** shown in FIG. 1).

(194) As shown in FIG. 5I, a plurality of queues **530** may be placed in proximity (e.g., within reach) of a plurality of corresponding mechanical arms **542**. The mechanical arms **542** may be the same as, or similar to, the mechanical arms described above with respect to FIGS. 1-5 (e.g., the mechanical arm **202** shown in FIG. 2A). The mechanical arms **542** may each have a gripper (e.g., the gripper **400**) configured to transfer slides between the queues **530** and individual slide processing stations **544**. The slide processing stations **544** may be the same as, or similar to, the slide processing stations described above (e.g., the slide processing stations **210**), and may be configured to perform one or more processing or manufacturing operations on slides during a pathology slide manufacturing process.

(195) Additionally, one or more slide carriers **550** may be positioned on a work station surface or substrate **552**, and may be accessible by (e.g., within reach of) the gripper of the mechanical arm **542**. The slide carriers **550** may be the same as, or similar to, the slide carriers described above (e.g., the slide carriers **208**, **214a**, **214b**, and **300**).

(196) Utilizing the cylindrical or rotary queue **530** in the slide processing system **540** may enable mechanical arms **542** to be located in a single location that is capable of reaching various components of the slide processing system **540**, including the queues **530**, and the various slide processing stations **544** and the slide carriers **550**. Additionally, as shown in FIG. 5I, the slide processing system **540** may be modular and enable additional work stations **552**, with additional mechanical arms **542**, queues **530**, and slide processing stations **544** to be added to or removed from the environment of the slide processing system **540** as the demands on the slide processing system **540** vary over time.

(197) FIG. 5J is a flow diagram illustrating a method of manufacturing a slide in a slide processing system, according to some embodiments. Although various operations are illustrated in FIG. 5J as part of the method of manufacturing a pathology slide in a slide processing system, embodiments according to the present disclosure are not limited to the operations illustrated in FIG. 5J. For example, some embodiments may include additional operations or fewer operations without departing from the spirit and scope of embodiments according to the present disclosure.

(198) As described above, the manufacturing process starts, and then, at operation **570**, a gripper may grip a slide in a first orientation (e.g., any of the orientations described above with respect to FIGS. 3A-4M). At operation **572**, the gripper may insert the slide into a slot of a slide carrier cassette (e.g., the slide carrier cassette **500**). At operation **574**, the gripper may place the slide carrier cassette into a queue having a cavity corresponding to a shape of the slide carrier cassette. At operation **576**, the gripper may remove the slide from the slide carrier cassette. At operation **578**, the gripper may place the slide into a component of the slide processing system to enable a slide manufacturing operation to be performed on the slide.

(199) Thus, as shown above, embodiments according to the present disclosure enable a relatively improved and efficient pathology slide manufacturing process. For example, during a pathology slide manufacturing process, a mechanical arm with a gripper may be utilized to manipulate and transfer the slide between different operations and locations. Different slide processing components, stations, or carriers, however, may require that slides be picked up or inserted from various angles, locations, and orientations. Thus, embodiments according to the present disclosure may include a slide carrier may be provided that enables a mechanical arm gripper to pick up and set down a slide during intermediate operations of a slide manufacturing process. The slide carrier may enable the gripper to pick up the slide in various modes or configurations according to the

processing operation to be performed next. For example, the slide may be placed onto the slide carrier using a grip with one orientation, and then the gripper may pick the slide up using a grip with a different orientation. Because of various angles of the surface of the slide carrier relative to gravity, the slide may tend to move or gravitate toward a known location on the slide carrier, thereby enabling the gripper and mechanical arm to pick up the slide again from a known location and orientation.

(200) Additionally, embodiments according to the present disclosure may include a gripper or gripper fingers having various protrusions, channels, or grooves, to enable a pinching or compression action to be performed on the slide from different angles or orientations.

(201) Finally, some embodiments may include a slide carrier cassette configured to hold a plurality of slides, and to register slides at a known location after being inserted into the slide carrier cassette. A plurality of slide carrier cassettes may be inserted into a queue to be accessed by a mechanical arm and gripper system. In some embodiments, the queue may have a cylindrical or rotary configuration, enabling individual slide carrier cassettes or individual slides to be moved to an accessible location of a mechanical arm during a pathology slide manufacturing process.

(202) As described above with respect to FIG. 1, the pathology environment is generally labor and time intensive. Generally speaking, alternative systems have relatively little automation and involve numerous individuals and operations. Highly trained pathologists and histotechnicians/histotechnologists are responsible for executing a large number of repetitive, manual, skilled, and unskilled tasks. Each task, when performed by a human, adds costs to the process, and creates the potential for errors. Additionally, as the qualified labor force is shrinking in many countries, it may be desirable to automate aspects of the pathology environment/histology process in order enable workers to direct their attention toward more complex tasks and problems.

(203) Thus, aspects of some embodiments may include an automated histology system (e.g., the slide processing system **200** illustrated and described, for example, with respect to FIG. 2A) capable of receiving pathology slides and tissue samples, and processing the slides or samples, for example, by way of one or more staining operations. According to some embodiments, as described in more detail below, the automated histology system may be capable of processing a large volume of pathology slides without human involvement.

(204) According to some embodiments, an automated histology storage system and method may enable consolidating pathology slides into an efficient vertical storage solution, which may reduce the overall footprint of the storage of pathology slides. According to some embodiments, the automated histology system may be capable of electrically connecting to a database, memory system, or network that stores information regarding tissue samples and patients in order to efficiently and automatically assign slides to appropriate cases, folders, or trays without human involvement. Additionally, according to some embodiments, the automated histology system may be capable of storing slide locations in a database, memory system, or network-connected memory in order to prevent or reduce lost time due to humans manually searching for errant slides. Some embodiments may further be capable of sorting pathology slides into cases, folders, or trays using a combination of one or more robotic arms and grippers, thereby reducing or eliminating the need for human involvement.

(205) FIGS. 6A-6D illustrate aspects of an example automated histology storage system and environment according to some embodiments. FIG. 6A illustrates a histology system and environment **600** according to some embodiments. According to some embodiments, the histology system and environment **600** may operate within, or as part of, a slide processing or manufacturing environment or system (e.g., the slide processing or manufacturing system **200**) As illustrated in FIG. 6A, the histology system and environment **600** may include an automated histology storage system **602**. As will be described in more detail below, the automated histology storage system **602** is configured to receive coverslipped slides at various stages of processing, and automatically organize and store the slides during the manufacture and processing of the slides within the

histology system and environment **600**.

(206) The histology system and environment **600** may further include one or more processing stations or devices **604a-604c**. The processing stations or devices **604a-604c** are configured to receive slides with tissue samples or specimens (e.g., coverslipped slides), and perform one or more staining or reagent operations on the slides as part of the manufacturing process of the slides in preparation for further analysis by a pathologist as described above. Although three processing stations or devices **604a-604c** are illustrated in FIG. **6A**, a person having ordinary skill in the art would recognize the histology system and environment **600** is not limited to three processing stations or devices, and according to some embodiments, the histology system and environment **600** may include additional, or fewer, processing stations or devices without departing from the spirit and scope of embodiments according to the present disclosure.

(207) A technician **606** may work within the histology system and environment **600** to initiate and monitor the processing operations of slides at the processing stations or devices **604a-604c**, and to transfer slides between the processing stations or devices **604a-604c**. The technician **606** may further transfer slides to the automated histology storage system **602** at various stages of the manufacturing process.

(208) FIG. **6B** shows further details of an example automated histology storage system **602** according to some embodiments. In particular, FIG. **6B** illustrates an example front perspective view of the automated histology storage system **602** at a side that may be accessible to a technician (e.g., the technician **606**). As will be described in more detail below, the automated histology storage system **602** operates as a mechanism to receive pathology slides and then, in an automated fashion, organize and store the slides, for example, between processing operations. Additionally, as will be described in more detail below, the automated histology storage system **602** may be capable of sorting stained slides, and assembling the slides into cases or folders for pathologist review, which is otherwise a labor and space intensive process that presents many opportunities for human error.

(209) As will be described in more detail below, slides may be input into the automated histology storage system **602** by way of one or more racks or trays, which a user or technician may load into an open position in an input rack portal. In some instances, the slides may have a tissue sample or specimen, which may be stained, dried, and coverslipped prior to being inserted into the automated histology storage system **602**. Embodiments according to the present disclosure are not limited thereto, however, and in some embodiments, the automated storage system **602** may be configured to receive slides at intermediate processing steps. Additionally, in some embodiments, one or more processing stations or devices (e.g., the processing stations or devices **604a-604c**) may be enclosed or located within the automated histology storage system **602**, such that one or more slide manufacturing processing operations may be executed by the automated histology storage system **602**.

(210) Individual slides may be labeled or marked with unique identifying information (e.g., in the form of a barcode or QR code). Once loaded into the automated histology storage system **602**, the slides may be individually scanned, and then loaded into a temporary storage location, for example, a slide carrier cassette **500** described above, to await being assembled into a case or folder that will be transferred to a different environment (e.g., a high resolution digital scanner or camera, or a pathologist) for further review and processing.

(211) When it is time to assemble a case or folder, a user or technician may initiate a folder creation process by providing a user input to the automated histology storage system **602**. The automated histology storage system **602** may then retrieve the relevant slides from one or more queues of slide carrier cassettes, and automatically assemble the case or folder with the relevant slides. The assembled case or folder may then be automatically transferred to an output portal to be retrieved by a user or technician.

(212) As shown in FIG. **6B**, the automated histology storage system **602** may include a plurality of

input portals **610** (shown and described in more detail below) to enable a user or technician to insert trays or racks of slides into the automated histology storage system **602**. That is, the input portals **610** operate as an interface for exchanging slides between users or technicians and the automated histology storage system **602**. According to some embodiments, the input portals **610** may be configured to receive slides stored, secured, or housed in temporary trays or racks as will be described in more detail below.

(213) In some embodiments, each portal may have the same width and height dimensions, and utilize the same mechanical and electrical connection interfaces with the automated histology storage system **602**. Accordingly, the input portals **610** may be configured to be modular components that can be interchanged (e.g., removed, replaced, etc.) according to the operational desires or requirements of the automated histology storage system **602**. For example, depending on the operations to be performed and the use case of the automated histology storage system **602**, it may be desirable to utilize a first type of portal (e.g., a tray portal, discussed in more detail below) for a period of time, and then remove the first type of portal so that a second type of portal (e.g., a rack portal, discussed in more detail below) can be inserted into the space or slot previously occupied by the first type of portal.

(214) The automated histology storage system **602** may further include one or more output portals **612** to enable the automated histology storage system **602** to provide one or more trays or racks of slides to a user or technician. That is, the output portals **612** may operate as an interface for users or technicians to retrieve trays or racks of slides from the automated histology storage system **602**.

(215) The automated histology storage system **602** may further include a user interface. According to some embodiments, the user interface may include a graphical display device or display panel **614** for display images or a graphical user interface. In some embodiments, the display panel **614** may include a touch screen capability to enable users to provide touch input. The automated histology storage system **602** may further include other mechanisms for providing user input, such as a keyboard, mouse, and one or more buttons.

(216) Using the user interface, a user or technician may provide an input to the automated histology storage system **602** to indicate that the user desires to deposit a slide, folder of slides, or rack of slides, into the automated histology storage system **602**. For example, the user or technician may provide identifying information about the slides, and the storage mechanism (e.g., whether the slides are in a folder, a rack, etc.). The automated histology storage system **602**, in some embodiments, may direct the user to a particular input portal **610** for depositing the slides. Once deposited into the input portal **610**, as will be described below, the automated histology storage system **602** may proceed to organize, sort, and assemble the slides into one or more storage queues until the user requests that they be retrieved.

(217) For example, when it is desirable to retrieve a slide, folder of slides, or rack of slides from the automated histology storage system **602**, a user or technician may provide an input to a user interface of the automated histology storage system **602** to request certain slides be retrieved. As will be described in more detail below, in response to a retrieval request, the automated histology storage system **602** may identify the corresponding slides, retrieve them from a storage queue, and insert the slides into one of the output portals **612**.

(218) According to some embodiments, the user interface may enable the user, for example, to request slides to be retrieved and/or assembled into one or more trays or racks. The slides may be stored in a variety of locations or slide carrier cassettes within the automated histology storage system **602**. For example, some of the requested slides may be stored in one location or slide carrier cassette and other slides may be stored in different locations or slide carrier cassettes. According to some embodiments, the user interface may enable the user to provide information about the characteristics of slides to be retrieved. For example, the user may be enabled to request all slides corresponding to a particular patient, all slides corresponding to a plurality of patients, or a subset of slides for a particular patient or group of patients that have certain characteristics (e.g.,

tissue sample from a certain part of the patient's body, or having certain stain or reagent characteristics, etc.). In response to the user input, the automated histology storage system **602** may proceed to retrieve relevant slides from the variety of storage locations within the automated histology storage system **602**, and assemble them into one or more trays or racks, while not including or retrieving any slides that do not fit the criteria of the request from the user.

(219) In some embodiments, the automated histology storage system **602** may further include one or more service doors **616**. In some embodiments the service doors **616** may be transparent (or have transparent regions) to enable a user or technician to observe the inside of the automated histology storage system **602**. Additionally, the service doors **616** may be configured to be opened in order to enable servicing or repair of components within the automated histology storage system **602**.

(220) FIG. **6C** shows further details of an example automated histology storage system **602** according to some embodiments. In particular, FIG. **6C** illustrates an example top front perspective view of the automated histology storage system **602** according to some embodiments. As illustrated in FIG. **6C**, the automated histology storage system **602** may include a mechanical or robotic arm **620**. The mechanical arm **620** may be the same as or similar to the mechanical arm **202** described above, and some repeated or duplicative description thereof may be omitted. The mechanical arm **620** may be configured to retrieve slides, slide folders, and slide racks from the input portals **610**, manipulate individual slides while organizing and assembling them in storage racks or queues, and then retrieve slides, folders, or racks from storage locations and deposit them in output portals **612**. The retrieval, manipulation, and storage operations of the mechanical arm **620** may be executed in response to user input (e.g., by way of the user input interface and the graphical display panel **614**), individual data corresponding to each slide, and one or predetermined handling protocols.

(221) The automated histology storage system **602** may further include one or more slide carrier queues **624**. The slide carrier queues **624** illustrated in FIG. **6C** may be the same as, or similar to, the slide carrier queues **530** illustrated and described above, and some repetitive or duplicated description thereof may be omitted. The slide carrier queues **624** may be capable of storing a plurality of racks (e.g., slide carrier cassettes **500**, in FIGS. **5A-5E**) of slides, as described above with respect to FIGS. **5F-5I**. Additionally, although FIG. **6C** illustrates the slide carrier queues **624** being configured in a planar configuration of rows and columns of cavities for storing slides, embodiments according to the present disclosure are not limited thereto, and in some embodiments, the slide carrier queues **624** may be configured in a cylindrical shape as shown in FIG. **5G**, and the cylindrical shape may be configured to be rotated, and different levels or rows may be modular in that they can be added or removed as desired.

(222) Additionally, according to some embodiments, the automated histology storage system **602** may include a tool change shelf or rack **630**. The tool change rack **630** may be configured to hold or store a plurality of different tools or grippers for use by the mechanical arm **620**. In some embodiments, the automated histology storage system **602** may be configured to automatically (e.g., without human involvement) change the tool or gripper being utilized by the mechanical arm **620** depending on the operation being performed. For example, a first type of gripper may be utilized for removing slide trays or racks (e.g., slide carrier cassettes) from an input portal **610**, but a different tool or gripper may be utilized for manipulating individual slides within the automated histology storage system **602**. Thus, in some embodiments, between operations with slides, the automated histology storage system **602** may send an electronic signal to the mechanical arm **620** to cause the mechanical arm **620** to remove a first tool or gripper and store it in the rack **630**, and then attach a second tool or gripper to the mechanical arm **620** that is stored on the rack **630**. As a person having ordinary skill in the art would recognize, embodiments according to the present disclosure are not limited to two grippers or tools, however, and any suitable number of tools or gripper configurations may be utilized according to the design and configuration of the automated histology storage system **602**.

(223) FIG. 6D shows an example top view of the automated histology storage system **602** according to some embodiments. As shown in FIG. 6D, the automated histology storage system **602** may include a working platform or workstation **640** to be utilized by the mechanical arm **620** to manipulate and process slides within the automated histology storage system **602**. According to some embodiments, various components may be fixed to the workstation **640** (or other locations within the automated histology storage system **602**). For example, according to some embodiments, the slide carrier **300** may be fixed to the workstation **640**. Additionally, the workstation **640** may include various sensors, for example, cameras, barcode readers, etc. for identifying aspects of components or slides within the automated histology storage system **602**. In some embodiments, the workstation **640** may further include one or more rack carriage nests and/or one or more tray carriage nests, which will be described in more detail below. The rack carriage nests and tray carriage nests may operate as intermediate platforms for enabling the mechanical arm **620** to work with and manipulate slides, trays of slides, and racks of slides. Further details of some components that may be mounted to the workstation **640** will be described below.

(224) As shown in FIG. 6D, the automated histology storage system **602** may further include or be electrically connected to a control system **650**. The control system **650** may include a processor **652** and a memory **654** coupled to (e.g., in electronic communication with) the processor **652**. The memory **654** may store instructions that, when executed by the processor **652**, cause the automated histology storage system **602** to execute various operations as described herein. For example, the instructions stored on the memory **654**, when executed by the processor **652**, may send one or more signals to the user interface to receive and display data or information to/from the user. Similarly, the instructions stored on the memory **654**, when executed by the processor **652**, may control or change the position or configuration of the mechanical arm **620** and/or the gripper of the mechanical arm **620** (for example, to pick up and manipulate slides, folders, trays, racks, etc.).

(225) The control system **650** may further be in electronic communication (by way of a wired or wireless communication channel) with a network **660** (e.g., a local area network (LAN), private wide area network (WAN), and/or public wide area network such as, for example, the Internet). In some embodiments, the control system **650** may be configured to receive or transmit signals to external components or computer systems that are also connected to, or in electronic communication with, the control system **650** by way of the network **660**.

(226) In some embodiments, a database **662** may be connected to the control system **650**, for example, through the network **660**. The database **662** may store data corresponding to unique identifying information about individual slides, groups of slides, patient data, tissue sample data, processing operation data, and the like. In some embodiments, the database **662** may be remote with respect to the automated histology storage system **602**, or the database **662** may be local or housed within the automated histology storage system **602**.

(227) FIGS. 7A-7P illustrate aspects of a portal system operating as part of the automated histology storage system **602** according to some embodiments. Within the histology system and environment **600**, various slide storage devices may be utilized to rest or store slides during processing.

(228) For example, for certain advanced staining processes or applications, slides may be stored in racks or baskets in which individual slides can be arranged in a vertical orientation and slid into individual slots sequentially in front or behind each other. The racks can then be utilized to transport slides to and from processing machines, or in some instances, may be inserted into processing machines to enable staining or reagent process operations to be performed.

(229) FIG. 7A shows example racks **702a-702e**, which are all generally capable of storing individual slides in a vertical orientation with the front and rear faces of the slides facing toward the adjacent slides in front or behind. As may be apparent, however, each of the slide racks **702a-702e** has a similar shape or configuration, yet each individual design has a different form factor. For example, different vendors or processing operations may produce or utilize racks with slightly different shapes or form factors for a variety of reasons.

(230) Thus, it may be difficult or time consuming for an automated histology storage system **602** utilizing a mechanical arm **620** to pick up and manipulate a variety of different racks with different form factors, since each different form factor may have different locations, handles, or protrusions at which a gripper of the mechanical arm **620** may grip the racks.

(231) As a person having ordinary skill in the art would recognize, the example rack form factors illustrated in FIG. 7A is not intended to be limiting, and various other form factors may be utilized. Rather, the examples illustrated in FIG. 7A are merely intended to convey the concept that embodiments according to the present disclosure can accommodate a wide variety of a plurality of different rack form factors.

(232) Similarly, in other staining processes or applications, slides may be stored in planar folders or trays, where individual slides may be arranged, for example, in rows or columns and the front and back faces are co-planar with each other. Like the racks described above, the trays of slides can be utilized to transport slides to and from processing machines.

(233) FIG. 7B shows example slide trays or folders **704a-704g**, which are all generally capable of storing individual slides in a planar orientation, for example, with slides arranged in rows and columns and in which front and back faces of slides are co-planar. As may be apparent, however, each of the trays **704a-704g** has a similar shape or configuration, yet each individual design has a different form factor. For example, similar to the racks discussed above, different tray manufacturers may utilize different form factors for different applications for a variety of reasons.

(234) Thus, as is the case with the varying shapes of racks, the varying shapes of the trays may make it difficult for an automated histology storage system **602** using a mechanical arm **620** to pick up and manipulate a variety of different trays that all have different form factors, since the locations for picking up and manipulating the trays with the gripper of the mechanical arm **620** may vary between different form factors.

(235) As a person having ordinary skill in the art would recognize, the example tray form factors illustrated in FIG. 7B is not intended to be limiting, and various other form factors may be utilized. Rather, the examples illustrated in FIG. 7B are merely intended to convey the concept that embodiments according to the present disclosure can accommodate a wide variety of a plurality of different tray form factors.

(236) Thus, as will be described below, embodiments according to the present disclosure provide mechanisms to enable the automated histology storage system **602** to receive, pick up, manipulate, and store slides from a wide variety of racks and trays that may have different form factors, without necessarily requiring the mechanical arm **620** to change gripping mechanisms for each varying form factor.

(237) In the present disclosure and claims, the term “slide holder” may be utilized to refer collectively to either slide trays or slide racks. That is, a “slide holder” according to the present disclosure and claims refers to a component that is configured to physically hold slides in slide slots or cavities, either in a vertical orientation (as in the case of slide racks illustrated and described with respect to FIG. 7A) or a planar orientation (as in the case of slide trays or folders illustrated and described with respect to FIG. 7B).

(238) FIG. 7C shows example input or output portals according to some embodiments. As described with respect to FIGS. 6A to 6D, the input and output portals **610** and **612** may operate as an interface mechanism to enable users or technicians to insert slides into the automated histology storage system **602**. As shown in FIG. 7C, the automated histology storage system **602** may include a tray portal **710a**, which may be the same as or similar to the input portal **610** and/or the output portal **612** described above with respect to FIGS. 6A to 6D. The tray portal **710a** may be capable of receiving slide trays (e.g., which are the same as or similar to the trays **704a-704g** described above).

(239) Similarly, the automated histology storage system **602** may include a rack portal **710b**, which may be the same as or similar to the input portal **610** and/or the output portal **612** described above

with respect to FIGS. 6A to 6D. The rack portal **710b** may be capable of receive slide racks (e.g., which are the same as or similar to the racks **702a-702f** described above).

(240) According to some embodiments, the tray portal **710a** and the rack portal **710b** may have the same or similar external dimensions. For example, the tray portal **710a** and the rack portal **710b** may have the same or similar height H, width W, and/or depth D. Additionally, according to some embodiments, the tray portal **710a** and the rack portal **710b** may have the same or similar mechanical mounting hardware for securing the portals to the automated histology storage system **602**. Furthermore, according to some embodiments, the tray portal **710a** and the rack portal **710b** may have the same or similar electrical connection mechanisms for being electrically connected to the automated histology storage system **602**.

(241) Accordingly, the portals may operate as modular components of the automated histology storage system **602** to enable the portals to be swapped in or out of the automated histology storage system **602** according to the design, operation, or use case. For example, in some environments or use cases, it may be desirable to utilize only or primarily tray portals **710a**. In other environment or use cases, it may be desirable to utilize only or primarily rack portals **710b**. In some environments, it may be desirable to be able to change the configuration or number of different types of portals. Because the portals **710a** and **710b** may have the same or similar dimensions, it may be possible to configure or reconfigure the portals, and swap portals in and out, according to the design and operation of the automated histology storage system **602**.

(242) FIG. 7D shows further details of an example tray portal **710a** according to some embodiments. As shown in FIG. 7D, the tray portal **710a** includes four layers, tray levels, or portal slots although embodiments according to the present disclosure are not limited thereto. For example, in some embodiments, there may be additional layers or fewer layers without departing from the spirit and scope of embodiments according to the present disclosure. As shown in FIG. 7D, each level, or portal slot, may be configured to hold a tray carriage. Further details of the tray carriages will be described in more detail below, but each tray carriage may be configured to receive a tray of slides. Each tray carriage may have substantially the same external form factor such that the mechanical arm **620**, and its gripping mechanism, can pick up and manipulate the tray carriage in substantially the same manner without requiring adjustment of gripping positions or gripper tools. As will be described in more detail below, each tray carriage may further have a cavity configured to receive a tray of slides. The interior form factor of the cavity may vary according to the design of the tray. Thus, while each of the different variety of trays may have a slightly different shape or form factor, the interior of the cavity of the tray carriages may conform to the shape of their corresponding tray. According to some embodiments, the interior cavity of the tray carriages may be keyed to their corresponding tray type, such that only they tray design corresponding to the individual tray carriage can fit properly in the interior cavity, and other tray designs cannot properly fit or be fully seated within the interior cavity of the wrong tray carriage.

(243) By way of example, 7D illustrates a portal with three tray carriages **716a-716c** in the bottom three layers or portal slots **714b-714d**, and the top portal slot **714a** is empty. The bottom tray carriage **716a** includes a tray inside the internal cavity of the tray carriage **716a**, while the other two tray carriages **716b** and **716c** are empty or vacant, meaning no trays are inside their respective interior cavities. As a person having ordinary skill in the art would recognize, the particular configuration illustrated in FIG. 7D is only for purposes of illustration, and depending on the usage and circumstances of the automated histology storage system **602**, various portal slots and tray carriages may be occupied or vacant at various times.

(244) As shown in FIG. 7D, the tray portal **710a** may include a door **718** that can be opened to enable a user to place a tray into the portal **710a**. In some embodiments, the user may first interact with the user interface (e.g., the display panel **614** shown in FIGS. 6A-6D) prior to inserting a tray into the portal **710a**. For example, user interface may have a graphical interface for enabling the user to indicate that the user wishes to insert a tray into the automated histology storage system

602.

(245) The automated histology storage system **602**, in some embodiments, may be configured to sense whether or not any of the portals **710a**, or portal slots or levels within one or more of the portals **710a**, is currently vacant (e.g., does not currently have a tray inserted). For example, according to some embodiments, the automated histology storage system **602** may include one or more sensors (e.g., magnets, visual or non-visual wavelength light sensors, cameras, etc.), configured to detect whether or not a portal slot has a tray carriage, and whether or not a tray carriage is occupied by a tray.

(246) In response to identifying an open portal or portal slot, the automated histology storage system **602** may identify whether or not a portal slot currently has a tray carriage inserted that is not occupied (i.e., is vacant) by a tray. If the portal slot is empty (i.e., does not have a tray or a tray carriage inserted), the automated histology storage system **602** may transmit one or more signals to the mechanical arm to retrieve an empty tray carriage and insert the empty tray carriage into the empty portal slot. Prior to inserting the empty tray carriage into the empty slot, however, the automated histology storage system **602** may prompt the user to identify the type of tray to be inserted into the portal **710a**. For example, the user may select the type of tray from a menu, or the user may provide unique identifying information about the tray (e.g., in the form of a barcode scan, etc.), to enable the automated histology storage system **602** to identify which tray carriage to select that will correspond to the tray to be inserted. In other embodiments, the automated histology storage system **602** may automatically identify the type of tray inserted into the portal **710a**. For example, each portal may include a barcode reader, camera, RFID reader, NFC scanner or other device that is able to retrieve unique identifying information about the tray, including the type and interior and exterior form factors of the tray.

(247) Similarly, if the portal **710a** has portal slots with vacant tray carriages, but the vacant tray carriages do not correspond to the tray that the user wishes to insert into the portal **710a**, the automated histology storage system **602** may be configured to sense, by way of one or more sensors, data stored in memory, and/or user input, that one of the tray carriages needs to be replaced with a tray carriage that matches the tray that the user wishes to insert. Accordingly, the automated histology storage system **602** may transmit one or more signals to the mechanical arm **620** to cause the mechanical arm **620** to remove one or more of the vacant tray carriages from the portal **710a**, retrieve a tray carriage with an inner cavity that corresponds to the tray that the user desires to insert into the portal **710a**, and insert the corresponding tray carriage into a vacant slot in the portal **710a**.

(248) Once the vacant tray carriage is ready to be filled with a tray, according to some embodiments, the automated histology storage system **602** may provide or display a graphical alert to the user regarding the availability and/or location of the proper vacant tray carriage. For example, in some embodiments the automated histology storage system **602** may display a message on a graphical user interface displayed on the display panel **614** to direct the user to the appropriate portal and portal slot. In some embodiments, the automated histology storage system **602** may display a light or other indicator at a portal slot.

(249) As shown in FIG. 7D, tray carriages are configured to be moved laterally in and out of the portal **710a**. According to some embodiments, however, the user may be prevented from entirely removing the tray carriage from the portal slot, and rather the tray carriage may only slide out of the portal to a fixed or predetermined limit that provides sufficient clearance to enable the user to insert the tray into the interior cavity of the tray carriage without removing the tray carriage. The mechanism for preventing the tray carriage from being entirely removed will be described in further detail below, but in some embodiments, the tray carriage may have one or more rails that are configured to be seated within and to slide along one or more slots or grooves, and a protrusion or peg at the back of the tray carriage may catch on the groove or the body of the portal **710a** to prevent the tray carriage from being pulled all the way out of the portal **710a**. Additionally, or

alternatively, the slots or grooves along which the tray carriage slides may not extend entirely to the front edge of the portal **710a**, but rather may be recessed within the portal **710a** by a predetermined distance in order to prevent the tray carriage from being pulled all the way out of the portal **710a**.

(250) FIG. 7E shows further details of internal components of a portal according to some embodiments. For example, FIG. 7E shows two layers or levels of portal slots of the tray portal **710a**, with the exterior housing and structure, and two of the four layers illustrated in FIG. 7D, removed. As shown in FIG. 7E, each layer or portal slot may be configured to accommodate a single tray carriage **716**. As described above, when inserted into the corresponding portal slot **714**, the tray carriages **716a** and **716b** are configured to slide in and out of the portal slot **714** to enable a tray to be inserted into the tray carriage. FIG. 7E illustrates an example in which different trays **720a** and **720b** are inserted into the tray carriages **716a** and **716b**, respectively. As shown in FIG. 7E, the trays **720a** and **720b** are similar to each other, in that they are both configured to receive or hold individual slides in a plurality of columns and rows, in which the front and rear faces of the slides are generally co-planar with each other when stored in their respective trays. But, each of the trays **720a** and **720b** have a slightly different exterior form factor. Thus, the tray carriage **716a** has an internal cavity configured to specifically accommodate the tray **720a**. Similarly, the tray carriage **716b** has an internal cavity configured to specifically accommodate the tray **720b**. If the tray **720a** is attempted to be inserted into the tray carriage **716b**, it will not fit or will not be seated properly, or there may be undesirable movement or spacing around the edges of the tray **720a**. Similarly, if the tray **720b** is attempted to be inserted into the tray carriage **716a**, it will not fit or will not be seated properly, or there may be undesirable movement or spacing around the edges of the tray **720b**.

(251) As shown in FIG. 7E, the tray carriages **716a** and **716b**, despite having different form factors in their respective interior cavities that correspond to specific tray designs, have the same or similar exterior form factor. Accordingly, the tray carriages **716a** and **716b** are capable of being positioned in any open portal slot. Each tray carriage **716a** and **716b** may include the same or similar exterior shapes or protrusions. For example, the tray carriages **716a** and **716b** may include one or more rails **724** for guiding the tray carriages **716a** and **716b** along slots or grooves **726** of the corresponding portal slot. Additionally, as described previously and further below, the tray carriages **716a** and **716b**, in some embodiments, may be configured to slide in and out of the tray portal **710a**, but may be prevented from being entirely removed from the tray portal **710a** from the front side (e.g., by way of a protrusion or slide mechanism that prevents the tray carriages **716a** and **716b** from being removed from the tray portal **710a**).

(252) FIG. 7F shows further details of the example tray carriages **716a** and **716b**. As described above, the tray carriages **716** operate as intermediate components or substrates for enabling a user to load and unload trays or folders of slides into the automated histology storage system **602**. The external features (e.g., gripping locations, handles, rails, grooves, dimensions, etc.) may be the same or similar, but a cavity for holding trays may vary according to the tray design corresponding to the tray carriage. The mechanical arm **620** of the automated histology storage system **602** may interact with the tray carriages **716**, rather than the tray itself. Because the exterior form factor is the same, the mechanical arm **620** of the automated histology storage system **602** may be capable of interacting with the different tray carriages **716** the same way, without having to change gripping tools or mechanisms.

(253) FIG. 7G shows a bottom view of a portion of a portal slot of the portal **710a** according to some embodiments, when a tray carriage **716a** is in an extended or pulled-out position. As shown in FIG. 7G, a protrusion or peg **728** may protrude or extend from a bottom surface or substrate of the portal slot. The protrusion **728** may be configured to fit within a groove or slot **729** on the bottom side of the tray carriage **716a**. According to some embodiments, the groove **729** may extend all the way to the front of side of the tray carriage **716a**, such that the tray carriage **716a** can be removed from the portal **710a** from the back side of the portal **710a**. But, the slot **729** may not

extend entirely to the back side of the tray carriage **716a**. Thus, as the tray carriage **716a** is moved in and out of the portal slot, the tray carriage **716a** may not be removed from the portal **710a** from the front side. That is, the protrusion may prevent the tray carriage **716a** from being pulled out past a predetermined point.

(254) FIG. 7H shows a close-up view of the rail **724** of an example tray carriage **716b** according to some embodiments. As shown in FIG. 7H, the rail **724** of the tray carriage **716b** may be configured to slide within the groove **726** to guide the tray carriage **716b** in and out of the portal slot **714c**. Additionally, the mechanical coupling of the rail **724** and the groove **726** operates to prevent a user from moving the tray carriage **716b** vertically.

(255) FIG. 7I shows a close-up view of a portion of the internal structure of a portal slot when a carrier tray is in an extended or pulled-out position. FIG. 7J shows a close-up view of the portion of the internal structure of the portal slot when the carrier tray is in a retracted or pushed-in position. As shown in FIG. 7I, the groove **726** guides the rail **724** of the tray carriage **716b** to move in and out within the tray portal **710a**. As shown in FIG. 7I, the rear of the groove **726** may be cut away to form a cutout portion **730**, such that the groove **726** is located at the front portion of the tray portal **710a** and not the rear portion. Thus, as shown in FIG. 7J, when the tray carriage **716b** is in the pushed-in or retracted position, the rail **724** is no longer engaged with the groove **726**. Thus, from the back or rear side (i.e., internal with respect to the automated histology storage system **602**) of the portal **710a**, the tray carriage **716b** may be lifted and removed from the portal **710a** by the mechanical arm **620** and pulled into the interior of the automated histology storage system **602**.

(256) FIG. 7K illustrates further details of internal components of a rack portal according to some embodiments. For example, FIG. 7K shows four rack slots **740a-740d** of the rack portal **710b**, with the exterior housing and structure removed. Although four rack slots are illustrated in FIG. 7K, embodiments according to the present disclosure are not limited thereto, and according to some embodiments, there may be fewer rack slots or additional rack slots without departing from the spirit and scope of embodiments according to the present disclosure. Additionally, as discussed above, the rack portal **710b** and the tray portal **710a** may have the same or similar exterior dimensions, such that the number of rack slots in the rack portal **710b** may be a function of the shared width of the rack portal **710b** and the tray portal **710a**, and may further be a function of the width of the largest or widest tray that the tray portal **710a** is designed to accommodate.

(257) As shown in FIG. 7K, each rack slot **740a-740d** may be configured to accommodate a single rack carriage **742**. As described above, when inserted into the corresponding portal slot **740**, the rack carriages **742** are configured to slide in and out of the portal slot **740** to enable a rack to be inserted into the rack carriage. FIG. 7K illustrates an example in which racks **744a-744d** are inserted into rack carriages **742a-742d**, respectively. The racks **744a-744d** may correspond to one of the racks **702a-702f** illustrated above.

(258) As shown in FIG. 7K, each of the racks **744a-744d** are the same as each other. But, as described above with respect to FIG. 7A, a variety of different racks, having a variety of different form factors, may be utilized as part of the histology system and environment **600**. Thus, like the tray carriages **716**, the rack carriages **742a-742d** may each have the same exterior form factor. But, the form factor of the internal cavity, configured to receive the racks **744**, may be vary according to the design and form factor of a corresponding rack **744**. For example, different racks **702a-702f** each have different form factors. A corresponding rack carriage **742** may be designed for each of the racks **702a-702f**, such that an individual rack can be properly seated in its corresponding rack carriage. But, each rack carriage **742** is keyed to the shape of its corresponding rack design, and thus racks that do not correspond to a rack carriage either cannot fit inside the wrong rack carriage or cannot be seated properly (e.g., fully seated, without lateral or vertical movement, etc.).

(259) FIG. 7L shows further details of example rack carriages **742** according to some embodiments. As described above, the rack carriages **742** operate as intermediate components or substrates for enabling a user to load and unload racks of slides into the automated histology

storage system **602**. The external features (e.g., gripping locations, handles, rails, grooves, dimensions, etc.) may be the same or similar, but a cavity for holding racks may vary according to the rack design or form factor corresponding to the rack carriage. The mechanical arm **620** of the automated histology storage system **602** may interact with the rack carriages **742**, rather than the rack itself. Because the exterior form factor is the same, the mechanical arm **620** of the automated histology storage system **602** may be capable of interacting with the different rack carriages **742** the same way, without having to change gripping tools or mechanisms.

(260) FIG. 7M shows an example rack carriage nest **750** according to some embodiments.

According to some embodiments, the rack carriage nest **750** may be located within the automated histology storage system **602**. For example, the rack carriage nest **750** may be mechanically fixed or secured to a surface of the workstation **640**. The rack carriage **750** may operate as a location and platform to enable the mechanical arm **620** to pick up slides from the rack **744** for further processing (e.g., placing slides in a slide carrier cassette **500** for storage in a slide carrier queue **530**, or placing slides into the rack **744**). According to some embodiments, the rack carriage nest **750** may have a surface on which a rack carriage **742** (having a rack **744** inserted therein) may be placed by the mechanical arm **620**. Like the slide carrier **300** illustrated, for example, in FIG. 3A, the surface of the rack carriage nest **750** may be inclined relative to an x-axis direction and a y-axis direction, similar to or the same as the slide platform **308** of the slide carrier **300** shown, for example, in FIG. 3A. Thus, some detailed description of the rack carriage nest **750** may be omitted for brevity. Because the surface of the rack carriage nest **750** is inclined with respect to the x-axis and y-axis, the rack carriage **742**, when placed on the rack carriage nest **750**, may gravitate to a known, fixed position and orientation, enabling the gripper of the mechanical arm **620** to pick up individual slides or the rack carriage **742** relatively easily. That is, like the slide carrier **300**, the rack carriage nest **750** may bias edges of the rack carriage **742** toward a known location (e.g., the bottom right or left corner or edge) of the rack carriage **750**.

(261) FIG. 7N shows an example tray carriage nest **760** according to some embodiments.

According to some embodiments, the tray carriage nest **760** may be located within the automated histology storage system **602**. For example, the tray carriage nest **760** may be mechanically fixed or secured to a surface of the workstation **640**. The tray carriage nest may operate as a location and platform to enable the mechanical arm **620** to pick up slides from the tray **720** for further processing (e.g., placing slides in a slide carrier cassette **500** for storage in a slide carrier queue **530**, or placing slides into the tray **720**). According to some embodiments, the tray carriage nest **760** may have a surface on which a tray carriage **716** (having a tray **720** inserted therein) may be placed by the mechanical arm **620**. In some embodiments, the top surface of the tray carriage nest **760** may be inclined relative to an x-axis and/or a y-axis to cause the tray carriage **716** to gravitate to a known, fixed position and orientation. Thus, the mechanical arm **620** may be capable of picking up individual slides or the tray carriage **716** relatively easily.

(262) FIGS. 7O and 7P illustrate aspects of a rack nest **770** according to some embodiments.

Certain types of staining and reagent machines may require the use of racks having certain form factors, as described previously. In some instances, the spacing between individual slides may be relatively tight. That is, in order to fit the desired number of slides into the rack while still allowing the rack to fit within the space parameters of the staining machine, each slide may be positioned relatively close to the slides in front and behind. Because of the tight spacing between slides, it may be difficult for the mechanical arm **620** to manipulate the gripper in a way that allows the gripper to fit into the tight spacing when trying to pick up individual slides, without bumping into adjacent slides.

(263) Thus, according to some embodiments, a rack nest **770** may be utilized to passively lift certain slides (e.g., every other slide) up to provide sufficient spacing to enable the mechanical arm **620** to grip the slides without contacting or damaging adjacent slides. FIG. 7O shows a side view of an example rack nest **770** according to some embodiments. FIG. 7P shows a perspective view of

the example rack nest **770** according to some embodiments. For example, according to some embodiments, each of the protrusions **774** has a height of approximately 1 inch.

(264) According to some embodiments, the mechanical arm **620** may place the rack **744** directly on the rack nest **770**, in a nesting region **772**. Within the nesting region **772**, the rack nest **770** may include one or more protrusions **774**. According to some embodiments, the protrusions **774** may be positioned at locations corresponding to every other slide or slide location within the rack **744**. According to some embodiments, each protrusion may have a width that is less than a distance between the front face and rear faces of adjacent slides. Thus, when the rack **744** is placed into the nesting region **772**, slides that correspond to or align with protrusions **774** will be lifted or raised by the protrusions **774** above the slides adjacent to them. Additionally, each protrusion may have a height that is sufficient to enable the gripper of the mechanical arm **620** to grip the slide that is raised by the protrusion **774** without the gripper contacting or damaging the adjacent slides that are not raised. Furthermore, according to some embodiments, each protrusion may have a height that is sufficient to enable unique identifying information (e.g., barcodes) of the raised slides to be exposed above the slides that are not raised.

(265) According to some embodiments, the mechanical arm **620** may lift the rack **744** to rotate the rack **744** by 180 degrees before re-inserting the rack into the nesting region **772** to raise the other slides. For example, in a first position, the protrusions **774** may cause a first group of slides (e.g., the even-numbered slides) to be raised above a second group of slides. In a second position, in which the rack **744** is rotated 180 degrees relative to the first position, the protrusions **774** may cause a second group of slides (e.g., the odd-numbered slides) to be raised above the first group of slides. Additionally, according to some embodiments, rather than having fixed or stationary protrusions **774**, the protrusions may be capable of being dynamically raised or lowered (e.g., by use of actuators) thereby enabling the automated histology storage system **602** to dynamically raise or lower specific slides according to the operation of the mechanical arm **620**. Thus, the rack nest **770** may ease dimensional constraints of the gripper of the mechanical arm **620** and improve robustness and ease of picking up and manipulating slides.

(266) FIGS. **8A-8E** illustrate an example slide position detection system according to some embodiments. As described above, for example, with respect to FIGS. **7O** and **7P**, slides in a rack **744** may be very close to each other, and the tight spacing between adjacent slides may make it difficult for the mechanical arm **620**, and the gripper of the mechanical arm **620**, to pick up individual slides. Additionally, in many instances, an individual rack **744** may not be completely filled with slides. Thus, for each rack, the mechanical arm **620** may have to spend a certain amount of time identifying whether or not a slide is present at each location in the rack **744**. Over the course of a day, where a large volume of slides may be processed through the automated histology storage system **602**, the amount of time that the mechanical arm **620** devotes to check whether or not a slide is present in various positions within the numerous racks it processes may be substantial. If the mechanical arm **620** mis-identifies a position within the rack as being empty when in fact a slide is present, it may cause a threat to the slide and the automated histology storage system **602**, in that the slide may break, which may cause debris within the automated histology storage system **602** that could damage other components or contaminate the environment and other slides.

(267) Thus, according to some embodiments, the automated histology storage system **602** may include a slide position detection sensor **800** configured to enable the automated histology storage system **602** to identify whether or not a slide is present at various locations in a given rack **744**. According to some embodiments, the slide position detection sensor **800** may include a depth camera (e.g., a stereoscopic depth camera) including, for example, two or more image sensors **802**.

(268) According to some embodiments, the slide position detection sensor **800** may be placed in front of the rack **744** and/or a rack nest **750/770** at a predetermined distance D from the rack **744**. Although the rack nest **770** is illustrated in FIG. **8A**, embodiments are not limited thereto, and according to some embodiments, the rack nest **750** (e.g., a rack nest without the protrusions **774**)

may be utilized. For convenience of description, the rack nest **770** may be referenced in connection with the slide position detection sensor **800**, but embodiments according to the present disclosure are not limited thereto.

(269) Because the distance *D* between the rack **744** and the slide position detection sensor **800** is known and fixed, the distance between the slide position detection sensor **800** and each slide slot or slide position can also be known. Thus, according to some embodiments, the slide position detection sensor **800** may be configured to iterate through the known distance of each slide position and capture an image at each predetermined distance of each slide position or slide slot to identify whether or not a slide is present at that position.

(270) Because the distance from the front of the rack **744** to the first slide position is known, and the distance from the slide position detection sensor **800** is known, the distance from the slide position detection sensor **800** to the first slide position is also known. Additionally, the distance between each subsequent slide position may be a known distance. For example, according to some embodiments, the distance between each slide position may be the same (e.g., 4 millimeters (mm), 6.5 mm, etc.), although embodiments according to the present disclosure are not limited thereto, and in some embodiments the distance between slide positions may vary but is known or predetermined.

(271) In order to detect or determine whether or not a slide is located at a given slide position, the slide position detection sensor **800** is capable of taking an image focusing on a specific distance away from the slide position detection sensor **800** that corresponds to the given slide position. If an element of a slide (e.g., a barcode, unique identifying information, etc.) is visible or identified at the specific distance, the slide position detection sensor **800** may determine that the slide position is occupied by a slide. By contrast, if an element of a slide is not visible or identified at the specific distance, the slide position detection sensor **800** may determine that the slide position is vacant (e.g., not occupied by a slide).

(272) For example, FIG. **8B** illustrates an example image of a first slide position (e.g., slide slot **1**) in a rack **744** that is a first distance away from the slide position detection sensor **800**. The first distance corresponds to the distance *D*, plus the distance from the front of the rack **744** to the first slide position (e.g., the front of the first slide position). In the example illustrated in FIG. **8B**, the left image **810** shows a depth map image or rendering of the slide rack **744**, and the right image **812** shows an RGB image of the slide rack **744**. An element **814** (e.g., a barcode) affixed to the surface of the slide is present at the first distance. Accordingly, the slide position detection sensor **800** and/or the automated histology storage system **602** may identify a slide is present at the first slide position, and may cause the mechanical arm **620** to interact with the slide (e.g., by picking the slide up, etc.).

(273) Because the slide rack **744** is positioned in the rack carriage **770**, the protrusions **774** (illustrated, for example, in FIGS. **7O** and **7P**) cause the slide in the second slide position, behind the slide in the first slide position, to be elevated above the slide in the first slide position (as described previously). Thus, if desired, prior to the slide in the first slide position being removed, the slide position detection sensor **800** may be configured to adjust the distance it is measuring/evaluating to the distance of the second slide position to identify whether or not an element of a second slide in the second slide position is visible.

(274) For example, FIG. **8C** illustrates an example image of a second slide position (e.g., slide slot **2**) in the rack **744** that is a second distance away from the slide position sensor **800**. The second distance corresponds to the distance *D*, plus the distance from the front of the rack **744** to the second slide position (e.g., the front of the second slide position). In the example illustrated in FIG. **8C**, the left image **820** shows a depth map image or rendering of the slide rack **744**, and the right image **822** shows an RGB image of the slide rack **744**.

(275) As shown in FIG. **8C**, the second slide is positioned behind the first slide illustrated in FIG. **8B**. An element **824** (e.g., a barcode) affixed to the surface of the slide is present at the second

distance. Accordingly, without the first slide being removed, the slide position detection sensor **800** and/or the automated histology storage system **602** may identify a slide is present at the second slide position, and may cause the mechanical arm **620** to interact with the slide (e.g., by picking up the slide, etc.).

(276) The automated histology storage system **602** and/or the slide position detection sensor **800** may continue iterating through each slide position of the rack **744** in order to help identifying whether or not each slide position is occupied by a slide. As slides are identified, they may be removed from the rack **744** in order to enable the slide position detection sensor **800** to view the next slide position. In instances where an element of a slide is not identified at a given slide position, the slide position detection sensor **800** and/or the automated histology storage system **602** may determine that no slide is present at that slide position.

(277) Alternatively, according to some embodiments, as will be described in more detail below, the slide position detection sensor **800** may be configured to capture one or more images of a slide rack **744** and identify a slide element is present in the image. Based on the slide element (e.g., a barcode, QR code, etc.) being present in the image, and a depth map or depth image information, the slide position detection sensor **800** may be configured to identify a distance between the slide position detection sensor **800** and the identified slide element. The slide position detection sensor **800** and/or the automated histology storage system **602** may then infer or identify (e.g., using a lookup table or stored distance data) to determine a slide position of the identified slide element.

(278) Thus, embodiments according to the present disclosure may enable the automated histology storage system **602** to automatically identify (e.g., without human involvement or intervention) the presence or lack of slides at individual slide positions without potential risks to the safety of slides and the environment of the automated histology storage system **602** that may otherwise be caused, for example, by the mechanical arm **620** attempting to grip slides that are not present or incorrectly determining a slide is not present when it is. Additionally, the slide position detection sensor **800** may enable relatively faster identification of whether or not a slide is present at each slide position.

(279) FIG. **8D** illustrates an example method to determine one or more slide positions according to some embodiments of the present disclosure. Although various operations are illustrated in FIG. **8D**, embodiments according to the present disclosure are not limited to the particular operations illustrated in FIG. **8D**. For example, according to some embodiments, there may be additional operations or fewer operations without departing from the spirit and scope of embodiments according to the present disclosure. Additionally, according to some embodiments, the order of operations may vary, unless otherwise stated or implied, without departing from the spirit and scope of embodiments according to the present disclosure.

(280) The operations illustrated in FIG. **8D** may be initiated and/or controlled by various elements of the automated histology storage system **602** including, for example, the control system **650**, the slide position detection sensor **800**, the mechanical arm **620**, etc. Unless otherwise stated, any suitable components within the automated histology storage system **602** may be utilized to execute the operations described in FIG. **8D**, and for convenience of description, the various components of the automated histology storage system **602** that perform the operations of the process illustrated in FIG. **8D** may be collectively referred to as “the system.”

(281) As illustrated in FIG. **8D**, the process starts and then, at operation **840**, the system initiates a slide position detection process. For example, according to some embodiments, the automated histology storage system **602** may detect the presence of a rack and/or a rack (e.g., the rack **744** and/or the rack carriage **742**) in a rack nest (e.g., the rack nest **750** or **770**). For example, according to some embodiments, a sensor (e.g., an optical sensor, a magnetic sensor, a mechanical switch, etc.) may be triggered in response to the rack and/or rack carriage being placed in the rack nest. Alternatively (or additionally), according to some embodiments, the automated histology storage system **602** and/or the mechanical arm **620** may identify the completion of an operation in which the mechanical arm **620** placed the rack and/or the rack carriage in the rack nest.

(282) In some instances, the rack and/or the rack carriage may be transported directly from an input portal to the rack carriage nest by the mechanical arm **620**, and the automated histology storage system **602** may not have information about which slots within the rack are occupied by slides. Thus, prior to manipulating (e.g., removing, processing, etc.) slides from the rack, the automated histology storage system **602** may identify or determine whether or not a slide is present in a given slide position within the rack, thereby enabling the mechanical arm **620** to safely begin manipulating the slide(s).

(283) Thus, at operation **842**, the system may capture one or more images of the slide rack and/or one or more exposed slides. According to some embodiments, the system may capture unique identifying information (e.g., in the form of a barcode, ID number, QR code, etc.) corresponding to the rack. For example, unique identifying information may be affixed (e.g., with a sticker or label) to the front face of the rack that is facing the slide position detection sensor **800**. The unique identifying information may identify information about samples on slides within the rack (e.g., patient information, tissue/specimen information, etc.), as well as the type or style of rack. For example, in some embodiments, different styles or types of slide racks may have different spacing between slide positions.

(284) According to some embodiments, the system may crop the one or more photos to correspond to a predetermined region at and around the rack, in order to reduce errors in subsequent processing.

(285) Based on a distance (e.g., a set or predetermined distance) from the slide position detection sensor **800** and the first slide position within the rack, the slide position detection sensor **800** may capture a color image of the rack and/or any exposed slides as well as a depth map image of the rack and/or any exposed slides, with the color and depth map images focusing on the known distance to the first slide position.

(286) At operation **844**, the system may determine whether or not a slide element is present in the color image and/or the depth map image at the distance to the first slide position. The slide element may include, for example, unique identifying information affixed to the slide, for example, in the form of a barcode, a QR code, identifying numbering or characters, and the like. If a slide element is present, the system may proceed, at operation **846**, to control the mechanical arm **620** to manipulate the slide in the first slide position, for example, by removing the slide and placing the slide into a slide carrier cassette **500**.

(287) The system may then proceed, at operation **848**, to determine whether or not the slide position is the last slide position within the rack. For example, based on the type or style of rack, the automated histology storage system **602** may know the number of slide positions within the rack, as well as the distance between the slide position detection sensor **800** and the final slide position (e.g., the slide position furthest from the slide position detection sensor **800**). If the slide position is the last slide position in the rack, then the process may end. Otherwise, if there are more slide positions in the rack, the system may return to operation **842**, to capture images corresponding to the next slide position.

(288) Alternatively, if a slide element is not present or identified at operation **844**, the system may proceed to operation **848**, to determine whether or not the vacant slide position is the last slide position in the rack. If the vacant slide position is also the last slide position in the rack, the process may end. If the vacant slide position is not the last slide position in the rack, the process may return to operation **842** to capture images corresponding to the next slide position.

(289) Additionally, or alternatively, according to some embodiments, rather than iterating through each individual slide position, according to some embodiments, the automated histology storage system **602** may capture images of slides and then when a slide is identified as being present, the corresponding slide position may be identified. For example, FIG. **8E** illustrates an example method to determine one or more slide positions according to some embodiments of the present disclosure.

(290) Although various operations are illustrated in FIG. 8E, embodiments according to the present disclosure are not limited to the particular operations illustrated in FIG. 8E. For example, according to some embodiments, there may be additional operations or fewer operations without departing from the spirit and scope of embodiments according to the present disclosure. Additionally, according to some embodiments, the order of operations may vary, unless otherwise stated or implied, without departing from the spirit and scope of embodiments according to the present disclosure.

(291) The operations illustrated in FIG. 8E may be initiated and/or controlled by various elements of the automated histology storage system **602** including, for example, the control system **650**, the slide position detection sensor **800**, the mechanical arm **620**, etc. Unless otherwise stated, any suitable components within the automated histology storage system **602** may be utilized to execute the operations described in FIG. 8E, and for convenience of description, the various components of the automated histology storage system **602** that perform the operations of the process illustrated in FIG. 8E may be collectively referred to as “the system.”

(292) As shown in FIG. 8E, the process starts and then, at operation **860**, the system initiates a slide position detection process. For example, according to some embodiments, the automated histology storage system **602** may detect the presence of a rack and/or a rack carriage (e.g., the rack **744** and/or the rack carriage **742**) in a rack carriage nest (e.g., the rack carriage nest **750** or **770**). For example, according to some embodiments, a sensor (e.g., an optical sensor, a magnetic sensor, a mechanical switch, etc.) may be triggered in response to the rack and/or rack carriage being placed in the rack carriage nest. Alternatively (or additionally), according to some embodiments, the automated histology storage system **602** and/or the mechanical arm **620** may identify the completion of an operation in which the mechanical arm **620** placed the rack and/or the rack carriage in the rack carriage nest.

(293) As described previously, in some instances, the automated histology storage system **602** may not know which slide positions within a rack are occupied by a slide and which positions are vacant. Thus, at operation **862**, the system may capture one or more images of the slide rack and/or one or more exposed slides. According to some embodiments, the system may capture unique identifying information (e.g., in the form of a barcode, ID number, QR code, etc.) corresponding to the rack. For example, unique identifying information may be affixed (e.g., with a sticker or label) to the front face of the rack that is facing the slide position detection sensor **800**. The unique identifying information may identify information about samples on slides within the rack (e.g., patient information, tissue/specimen information, etc.), as well as the type or style of rack. For example, in some embodiments, different styles or types of slide racks may have different spacing between slide positions.

(294) The one or more photos may include, for example, a color image (e.g., a high-resolution color image) of the rack and/or the one or more exposed slides, as well as a depth map image of the rack and/or the one or more exposed slides. According to some embodiments, the system may crop the one or more photos to correspond to a predetermined region at and around the rack, in order to reduce errors in subsequent processing.

(295) At operation **864**, the system may determine whether or not a slide element is present. For example, the system may identify whether or not unique identifying information corresponding to a slide (e.g., a barcode, QR code, identification numbers or characters, etc.) is present in the one or more photos. If a slide element is not present, the process may end.

(296) If a slide element is present, however, the system may proceed, at operation **866**, to identify the slide information. For example, the slide element may include unique identifying information about the patient, sample, and the like. The system may retrieve such information and store the information for subsequent processing.

(297) Additionally, at operation **868**, the system may create a pixel mapping between the color image and the depth image to identify coordinates of the slide element. Then, at operation **870**,

based on the coordinates of the slide element and the depth image, the system may identify the depth or distance of the slide element relative to the slide position detection sensor **800**. According to some embodiments, based on the distance derived from the coordinates and depth image, the system may identify the slide position of the slide element. For example, a lookup table (LUT) or slide position data may be stored in memory (e.g., the memory **652** and/or the database **662**) storing information about the distance between each slide position and the slide position detection sensor **800**. Once the slide position is identified, the system may cause the mechanical arm **620** to manipulate or move the slide out of the rack.

(298) Then, at operation **872**, the system may determine whether or not the slide was in the last slide position or not. If the position of the slide corresponds to the last slide position in the rack, the process may end. Alternatively, if the position of the slide does not correspond to the last slide position in the rack, the system may return to operation **862** to capture images corresponding to subsequent slide positions.

(299) FIGS. **9A-9E** illustrate a mechanical arm gripper according to some embodiments of the present disclosure. As described above with respect to FIGS. **2** and **3**, according to some embodiments of the present disclosure, a mechanical arm with a gripper connected to an end of the mechanical arm may be configured to pick up and manipulate a slide in a slide processing or manufacturing environment and/or an automated histology storage system.

(300) FIG. **9A** illustrates an example of a gripper **900** according to some embodiments. The gripper **900** may be the same as or similar to (or one example of) the gripper **216** illustrated in the slide processing or manufacturing system **200** of FIG. **2A**.

(301) The gripper **900** according to some embodiments may be configured to grip or manipulate a slide in various positions or orientations, in order to enable the slide to interface with various different carriages, racks, machines, and the like, in the process of manufacturing or storing a pathology slide. Thus, embodiments according to the present disclosure may reduce the need to change grippers or tools, and/or utilize multiple mechanical arms to manipulate slides.

(302) The gripper **900** may be formed of any suitable material that is solid enough to support a glass slide **204**, without a substantial risk of the glass slide or the gripper **900** being damaged. For example, in some embodiments, the gripper **900** may be formed of a plastic or polymer material, but embodiments according to the present disclosure are not limited thereto.

(303) As shown in FIG. **9A**, the gripper **900** may include two or more gripper fingers **902**. Although FIGS. **9A-9E** are described in the context of two gripper fingers **902a** and **902b**, embodiments according to the present disclosure are not limited thereto, and some embodiments may include additional gripper fingers **902** without departing from the spirit and scope of embodiments according to the present disclosure. In the present Specification, in some instances, the reference numeral **902** may be used to refer generally to each of the gripper fingers **902a** and **902b** (and any additional gripper fingers that may be utilized in some embodiments).

(304) Additionally, according to some embodiments, the gripper **900** (or the mechanical arm **620** to which the gripper **900** is attached) may include additional components configured to be utilized in connection with the manipulation and maneuvering of a slide during a slide manufacturing process. For example, in some embodiments, the gripper **900** (or the mechanical arm **620** to which the gripper **900** is attached) may include a suction cup for creating a suction force against a slide. Additionally, in some embodiments, the gripper **900** (or the mechanical arm **620** to which the gripper **900** is attached) may include a camera or sensor (e.g., an optical sensor, a pressure sensor, etc.) configured to identify or measure the location of a slide, a slide carrier, a surface, or the like.

(305) As shown in FIGS. **9A-9E**, the gripper **900**, using the gripper fingers **902a** and **902b**, may be configured to apply a pinching or inward force **F1** toward a center line **CENTER**, which bisects the gripper **900** and the slide **204**. The gripper fingers **902a** and **902b** may be positioned an equal distance from the center line **CENTER**, and may be configured to move toward and away from the center line **CENTER** in coordination with each other (e.g., at a same speed and with a same force).

(306) Referring to FIGS. 9A-9E, the gripper fingers **902a** and **902b** may be mounted to an actuator **904** configured to move the gripper fingers **902a** and **902b** inward (toward the center line CENTER) and outward (away from the center line CENTER) in order to open or close a space between the fingers **902a** and **902b**. The actuator **904** may be connected to a mechanical arm (e.g., the mechanical arm **202** shown in FIG. 2A and/or the mechanical arm **620** shown in FIG. 6A), such that collectively, the mechanical arm, the actuator **904**, and the gripper **900** may operate to manipulate a slide or move the slide from one location to another location. Each gripper finger **902a** and **902b** may have the same or similar shape. For example, in some embodiments, the gripper finger **902b** may be the same as the gripper finger **902a**, except rotated 180 degrees relative to the gripper finger **902a**, although embodiments according to the present disclosure are not limited thereto.

(307) FIG. 9A illustrates the gripper **400** holding a slide in a first gripping mode or configuration, in which the gripper fingers **902a** and **902b** operate to provide a pinching force **F1** widthwise (along an axis parallel to a width direction of the slide **204**), against the long edges of the slide **204**.

(308) In the first gripping mode, the gripper **900** may be configured to pick up and/or maneuver the slide **204** in an orientation in which the center line between the gripper fingers **902a** and **902b** extends along the front face of the slide **204** in parallel with the long edges of the slide **204**, with the center line bisecting the short edges of the slide **204**.

(309) The gripper fingers **902a** and **902b** may have a first portion **906** extending generally in a direction away from the actuator **904**. For example, according to some embodiments, the first portion **906** of the gripper fingers **902a** and **902b** may extend in a direction parallel (or generally parallel) to the center line CENTER.

(310) The gripper fingers **902a** and **902b** (e.g., the first portion **906** of the gripper fingers **902a** and **902b**) may have interior edges **908a** and **908b**, respectively, where the interior edges **908a** and **908b** are interior with respect to the opening or space between the gripper fingers **902a** and **902b**. According to some embodiments, the interior edges **908a** and **908b** may be elongated in a direction parallel to the center line CENTER and may define (or be located at) the interior edge the first portion **906** of each corresponding gripper finger **902a** and **902b**. In some embodiments, the interior edges **908a** and **908b** may extend parallel to each other, and may each have a groove therein, which are each adapted to receive the top (or bottom) of the long edges of a slide as shown in the first gripping mode shown in FIG. 9A. The groove in the interior edges **908a** and **908b** is illustrated and described in more detail below.

(311) When the pinching or inward force **F1** is applied to the long edges of the slide, the gripper **900** may be able to pick up the slide **204** and move the slide around in 3-dimensional space.

(312) FIG. 9B illustrates the gripper **900** holding a slide in a second gripping mode or configuration, in which the gripper fingers **902a** and **902b** operate to provide a pinching force or pressure **F2** lengthwise (along an axis parallel to a length direction of the slide **204**) against the short or width edges of the slide **204**. That is, the pressure or force **F2** may be in a direction toward (e.g., perpendicular to) the center line CENTER. In the second gripping mode, the gripper **900** may pick up and/or maneuver the slide **204** in an orientation in which the center line CENTER between the gripper fingers **902a** and **902b** may extend through the front and rear faces of the slide **204**, halfway between the short edges of the slide **204**. That is, a direction that is perpendicular or normal with respect to the planes of the front and rear faces of the slide **204** may be parallel to the center line CENTER, such that the front or rear face of the slide **204** faces toward the actuator **904**.

(313) The gripper fingers **902a** and **902b** may have a second portion **910** that extends from the first portion **906** at an angle parallel or generally parallel to the extension direction of the first portion **906**. The second portion **910** may include a protrusion or tooth **914** that extends toward the center line CENTER, such that a groove or cavity **916** is created between the protrusion **914** and the main body of the second portion **910**.

(314) The ends of the second portion **910**, including the groove **916**, are both further away from the

actuator **904**, and further away from each other, compared to the interior edges **908a** and **908b**. Thus, a relatively wider space may be created between the ends of the second portion **910** (and the grooves **916**), relative to the space between the interior edges **908a** and **908b**. Additionally, the groove **916** between the protrusion **914** and the main body of the second portion **910** may accommodate the width or short edges of the slide **204** such that the gripper **900** can pick up and/or maneuver the slide **204** with the front or rear face of the slide **204** facing toward the actuator **904**. (315) According to some embodiments, the gripper fingers **902a** and **902b** may be mechanically connected to the actuator **904** through support brackets **920a** and **920b**. That is, each gripper finger **902a** and **902b** may be connected to a corresponding support bracket **920a** and **920b**, respectively. According to some embodiments, the gripper fingers **902a** and **902b** may be mechanically affixed to the brackets **920a** and **920b** using any suitable mechanical connection mechanism. For example, in some embodiments, the brackets **920a** and **920b** may have a channel or groove **922** extending horizontally (e.g., perpendicularly with respect to the center line CENTER and/or parallel to the direction of the force F2). The support brackets **920a** and **920b** may have, for example, an upper portion (e.g., a flared upper portion) configured to fit within the channel **922**. Additionally, or alternatively, according to some embodiments, one or more screws or bolts may be inserted into one or more holes **924** that extend through a vertical portion of the support bracket **920**, and connect with one or more corresponding holes (e.g., threaded holes) located on the gripper fingers **902**. By utilizing the support brackets **920a** and **920b**,

(316) According to some embodiments, as illustrated for example in FIG. 9C, the gripper fingers **902a** and **902b** may include one or more notches or gaps. For example, a first notch **930** may be cut or machined into the interior edges **908a** and **908b**. The first notch **930** may be located at a first distance R1 from the bottom of the gripper finger (e.g., the outside edge of the protrusion **914**). Additionally, a second notch **932** may be cut or machined into the interior edges **908a** and **908b**. The second notch **932** may be located at a second distance R2, less than the first distance R1, from the bottom of the gripper finger. The first notch **930** and the second notch **932** may be utilized for manually calibrating pickup distances for the mechanical arm **620** when picking up slides. For example, a first pickup distance may be selected such that the width edge of the slide is aligned with the first notch **930** (as illustrated in FIG. 9C) when more precise control of the slide is desired. A second pickup distance may be selected such that the width edge of the slide is aligned with the second notch **932** when less precision is desired, but spacing (e.g., when a slide is located deep within a fixture, rack, or carriage) is a consideration. Additionally, the second notch **932** may operate to reduce overall friction on the slide at the edges **908a** and **908b**, because less surface area of the gripper fingers **902a** and **902b** may be in contact with the slide, which may reduce the likelihood of damage to the slide caused by the gripper fingers.

(317) Referring to FIGS. 9D and 9E, the interior edges **908** of the gripper fingers **902** may have a groove formed therein. According to some embodiments, the groove may have a rounded surface **940** for contacting the slide **204**, as illustrated in FIG. 9D. Alternatively, the groove may have a flat surface **940** for contacting the slide **204**. The rounded or flat surface **940** may be the inside surface of the groove that provides lateral pressure (e.g., in the direction of the force F1) against the long edges of the slide **204**. The rounded surface **940** may enable relatively improved grip position control. By contrast, the flat surface **940** may have relatively improved wear performance compared to the rounded surface, but may have slightly less grip control over the slide, since the slide might generally be capable of a greater degree of movement or angle inconsistency within the flat surface groove.

(318) FIGS. 10A-10F illustrate a mechanical arm gripper according to some embodiments of the present disclosure. As described above with respect to FIGS. 2, 3, 6, and 7, according to some embodiments of the present disclosure, a mechanical arm with a gripper connected to an end of the mechanical arm may be configured to pick up and manipulate a slide in a slide processing or manufacturing environment and/or an automated histology storage system. Additionally, a

mechanical arm equipped with a gripper may be configured to pick up and manipulate racks or trays that carry slides, as well as rack carriages or tray carriages that carry racks and trays, respectively.

(319) FIG. **10A** illustrates an example of a gripper **1000** according to some embodiments. The gripper **1000** may be the same as or similar to (or one example of) the gripper **216** illustrated in the slide processing or manufacturing system **200** of FIG. **2A**.

(320) The gripper **1000** according to some embodiments may be configured to grip or manipulate various carriages in various positions or orientations, in order to transfer slides or empty carriages to and from various locations within the slide processing or manufacturing system **200** and/or the automated histology storage system **602** the process of manufacturing or storing a pathology slide. Thus, embodiments according to the present disclosure may reduce the need to change grippers or tools, and/or utilize multiple mechanical arms to manipulate slides or carriages.

(321) The gripper **1000** may be formed of any suitable material that is solid enough to support a glass slide **204** and/or a carriage (e.g., a tray carriage **716**, a rack carriage **742**, etc.), without a substantial risk of the glass slide or the gripper **900** being damaged. For example, in some embodiments, the gripper **900** may be formed of a plastic or polymer material, but embodiments according to the present disclosure are not limited thereto.

(322) As shown in FIG. **10A**, the gripper **1000** may include two or more gripper finger brackets **1002**, each gripper finger bracket **1002** having one or more gripper finger rods **1004** protruding from their respective gripper finger brackets **1002**. Although FIGS. **10A-10F** are described in the context of two gripper finger brackets **1002a** and **1002b**, embodiments according to the present disclosure are not limited thereto, and some embodiments may include additional gripper finger brackets without departing from the spirit and scope of embodiments according to the present disclosure. In the present Specification, in some instances, the reference numeral **1002** may be used to refer generally to each of the gripper finger brackets **1002a** and **1002b** (and any additional gripper finger brackets that may be utilized in some embodiments).

(323) Additionally, although FIGS. **10A-10F** are illustrated and described in the context of each gripper finger bracket **1002** including two gripper finger rods **1004**, embodiments according to the present disclosure are not limited thereto, and some embodiments may include additional gripper finger rods or fewer gripper finger rods without departing from the spirit and scope of embodiments according to the present disclosure. In the present Specification, in some instances, the reference numeral **1004** may be used to refer generally to each of the gripper finger rods **1004a** and **1004b** (and any additional gripper finger rods that be utilized in some embodiments).

(324) Additionally, according to some embodiments, the gripper **1000** (or the mechanical arm **620** to which the gripper **1000** is attached) may include additional components configured to be utilized in connection with the manipulation and maneuvering of a slide during a slide manufacturing process. For example, in some embodiments, the gripper **1000** (or the mechanical arm **620** to which the gripper **1000** is attached) may include a suction cup for creating a suction force against a slide. Additionally, in some embodiments, the gripper **1000** (or the mechanical arm **620** to which the gripper **1000** is attached) may include a camera or sensor (e.g., an optical sensor, a pressure sensor, etc.) configured to identify or measure the location of a slide, a slide carrier, a surface, or the like.

(325) As shown in FIGS. **10A-10F**, the gripper **1000**, using the gripper finger brackets **1002a** and **1002b** and the gripper finger rods **1004a** and **1004b**, may be configured to apply a pinching or inward force **F3** toward a center of the gripper **1000**. The gripper finger brackets **1002a** and **1002b** may be positioned an equal distance from the center of the gripper **1000**, and may be configured to move toward and away from the center in coordination with each other (e.g., at a same speed and with a same force).

(326) Referring to FIGS. **10A-10F**, the gripper finger brackets **1002a** and **1002b** may be mounted to an actuator **1006** configured to move the gripper finger brackets **1002a** and **1002b** inward (toward the center) and outward (away from the center) in order to open or close a space between

the gripper finger brackets **1002a** and **1002b**. The actuator **1006** may be connected to a mechanical arm (e.g., the mechanical arm **202** shown in FIG. 2A and/or the mechanical arm **620** shown in FIG. 6A), such that collectively, the mechanical arm, the actuator **1006**, and the gripper **1000** may operate to manipulate a slide or move the slide from one location to another location. Each gripper finger bracket **1002a** and **1002b** may have the same or similar shape, and each gripper finger rod **1004a** and **1004b** may have the same or similar shape. For example, in some embodiments, the gripper finger bracket **1002b** may be the same as the gripper finger bracket **1002a**, except rotated 180 degrees relative to the gripper finger bracket **1002a**, although embodiments according to the present disclosure are not limited thereto.

(327) Referring to FIG. **10B**, each gripper finger rod **1004** may extend generally in a direction away from the gripper finger bracket **1002**. According to some embodiments, the gripper finger rod **1004** may have a generally cylindrical shape, with a consistent diameter along the length of the gripper finger rod **1004**. Embodiments according to the present disclosure are not limited thereto, and according to some embodiments, the gripper finger rod **1004** may have a rectangular or square shape (or any other suitable shape) in a cross-section, and in some embodiments, the gripper finger rod **1004** may have a curved shape in the extension direction, instead of a linear cylinder as illustrated in FIGS. **10A-10F**.

(328) The gripper finger brackets **1002** may be mounted to the actuator **1006** using any suitable connection mechanism, for example, one or more screws or bolts, clips, springs, etc. The gripper finger rods **1004** may be mounted to their corresponding gripper finger bracket **1002** by being inserted into a hole (e.g., a threaded hole), being connected to a threaded bolt extending from the gripper finger bracket **1002**, or any other suitable mechanism (e.g., a clip, spring, friction, adhesive material, etc.).

(329) According to some embodiments, the gripper finger rod **1004** may have one or more segments extending away from the gripper finger bracket **1002**. For example, FIG. **10B** illustrates the gripper finger rods **1004** having three segments **1008a**, **1008b**, and **1008c**, although embodiments according to the present disclosure are not limited thereto, and some embodiments may include fewer segments or additional segments without departing from the spirit and scope of embodiments according to the present disclosure.

(330) Additionally, one or more flared portions may be incorporated into the gripper finger rods **1004**. For example, FIG. **10B** illustrates the gripper finger rods having two flared portions **1010a** and **1010b**, although embodiments according to the present disclosure are not limited thereto, and some embodiments may include fewer flared portions or additional flared portions without departing from the spirit and scope of embodiments according to the present disclosure. According to some embodiments, the flared portions **1010** may be made of the same material as the segments **1008**. For example, the flared portions **1010** and the segments **1008** may be made of a polymer or plastic material, although embodiments are not limited thereto, and any other suitable material (e.g., metal) may be utilized. Additionally, according to some embodiments, the flared portions **1010** may be o-rings formed of a different material (e.g., rubber) from the segments, and may be installed or inserted into grooves or channels in the gripper finger rod **1004** corresponding to the location of the flared portions **1010**. The flared portions **1010** may operate to aid in friction when the gripper **1000** is being utilized to pick up objects (e.g., tray or rack carriages).

(331) Referring to FIGS. **10C** and **10D**, the rack carriage **742** (shown in FIG. **10C**) and the tray carriage **716** (shown in FIG. **10D**) may have one or more channels or grooves formed thereon at various locations. For example, as illustrated in FIG. **10C**, the rack carriage **742** includes a first channel **1020a** and a second channel **1020b** on a first side of the rack carriage **742**. On the opposite side of the rack carriage **742**, the rack carriage **742** also includes the first channel **1020a** and the second channel **1020b**. The first channel **1020a** may correspond, for example, to the first gripper finger rod **1004a** and the second channel **1020b** may correspond, for example, to the second gripper finger rod **1004b**. For example, the distance between the first channel **1020a** and the second

channel **1020b** may correspond to (or be equal to) the distance between the first gripper finger rod **1004a** and the second gripper finger rod **1004b** on the same gripper finger bracket **1002**.

Additionally, according to some embodiments, the shape of the channels **1020a** and **1020b** may have a contour that corresponds or conforms to the shape of the gripper finger rods **1004a** and **1004b**, respectively. Although FIG. **10C** illustrates the channels **1020** being at one end of the rack carriage **742**, embodiments according to the present disclosure are not limited thereto. For example, in various embodiments, the channels **1020** may be located at any suitable location, for example, the other end of the rack carriage **742**, the middle of the rack carriage **742**, etc.

(332) Similarly, referring to FIG. **10D**, the tray carriage **716** includes a first channel **1030a** and a second channel **1030b** at a top side of the tray carriage **716**. On the opposite side of the tray carriage **716**, the tray carriage **716** also includes two corresponding channels. The first channel **1030a** may correspond, for example, to the first gripper finger rod **1004a** and the second channel **1030b** may correspond, for example, to the second gripper finger rod **1004b**. For example, the distance between the first channel **1020a** and the second channel **1030b** may correspond to (or be equal to) the distance between the first gripper finger rod **1004a** and the second gripper finger rod **1004b** on the same gripper finger bracket **1002**. Additionally, according to some embodiments, the shape of the channels **1030a** and **1030b** may have a contour that corresponds or conforms to the shape of the gripper finger rods **1004a** and **1004b**, respectively. Although FIG. **10D** illustrates the channels **1030** being at a particular location (e.g., a handle) of the tray carriage **716**, embodiments according to the present disclosure are not limited thereto. For example, in various embodiments, the channels **1030** may be located at any suitable location around the tray carriage **716**.

(333) As shown in FIGS. **10E** and **10F**, the gripper **1000**, utilizing the gripper finger brackets **1002**, the gripper finger rods **1004** and the actuator **1006**, may be utilized to grip, manipulate, and maneuver the rack carriage **744** and the tray carriage **716** in three-dimensional space. Depending on the object being picked up or maneuvered, the gripper **1000** may rotate to an appropriate direction to apply the pinching force **F3** against the object. For example, as illustrated in FIG. **10E**, the force **F3** may be applied in a generally horizontal direction against opposite surfaces or edges of the rack carriage **744** in order to keep the rack carriage **744** upright and prevent a rack or slides from falling out of the rack carriage **744** during movement. The gripper **1000** may rotate (for example, by 90 degrees) in order to pick apply the force **F3** against opposite surfaces or edges of the tray carriage **716** in order to keep the tray carriage **716** upright and to prevent a tray or slides from falling out of the tray carriage **716** during movement.

(334) FIGS. **11A-11B** illustrate a mechanical arm gripper according to some embodiments of the present disclosure. As described above with respect to FIGS. **2**, **3**, **6**, and **7**, according to some embodiments of the present disclosure, a mechanical arm with a gripper connected to an end of the mechanical arm may be configured to pick up and manipulate a slide in a slide processing or manufacturing environment and/or an automated histology storage system.

(335) FIG. **11A** illustrates an example of a gripper **1100** according to some embodiments. The gripper **1100** may be the same as or similar to (or one example of) the gripper **216** illustrated in the slide processing or manufacturing system **200** of FIG. **2A**.

(336) The gripper **1100** may be configured to grip or manipulate a slide in various positions and orientations, in order to enable the slide to interface with various different carriages, racks, machines, and the like, in the process of manufacturing or storing a pathology slide. Thus, embodiments according to the present disclosure may reduce the need to change grippers or tools and/or utilize multiple mechanical arms to manipulate slides.

(337) The gripper **1100** may be formed of any suitable material that is solid enough to support a glass slide **204**, without a substantial risk of the glass slide or the gripper **1100** being damaged. For example, in some embodiments, the gripper **1100** may be formed of a plastic or polymer material, but embodiments according to the present disclosure are not limited thereto.

(338) As shown in FIG. **11A**, the gripper **1100** may include two gripper fingers **1102** and **1104** that

are configured to engage with each other to pick up a slide **204**. The first gripper finger **1102** and the second gripper finger **1104** may be configured to be coupled to an actuator of a mechanical arm (e.g., the actuator **904** in FIG. **9A** and/or the mechanical arm **620**) at mounting locations **1106** and **1108**, respectively.

(339) The gripper **1100**, using the gripper fingers **1102** and **1104**, may be configured to apply a pinching or inward force **F4** toward a center line CENTER, which bisects the slide **204**. The gripper finger **1102** may have a vertical portion **1110** extending in the Z-direction, and a horizontal portion **1112** extending in the Y-direction. Similarly, the gripper finger **1104** may have a vertical portion **1114** extending in the Z-direction, and a horizontal portion **1116** extending in the Y-direction. The horizontal portion **1112** of the first gripper finger **1102** may be configured to slide along (e.g., in contact with) the horizontal portion **1116** of the second gripper finger **1104**, as the vertical portions **1110** and **1114** move toward and away from each other when gripping or releasing a slide.

(340) FIG. **11B** shows a rear view of the first gripper finger **1102** when viewed in the Y-direction. As shown in FIG. **11B**, the first gripper finger **1102** may have a prong or protrusion **1120** that extends vertically below the horizontal portion **1112**. Similarly, referring to FIG. **11A**, the second gripper finger **1104** may have a prong or protrusion **1122** that extends vertically below the horizontal portion **1112** of the first gripper finger **1102**. Thus, as illustrated in FIG. **11A**, the protrusion **1120** of the first gripper finger **1102** and the protrusion **1122** of the second gripper finger **1104** may operate together to apply the pinching force **F4** against the slide **204** in a position in which the front or rear face of the slide **204** faces in the Z-direction.

(341) Additionally, an edge **1130** at the end of the horizontal portion **1112** of the first gripper finger **1102**, and an interior edge **1132** of the protrusion **1122** that faces the edge **1130**, may collectively operate together to apply the pinching force **F4** against the front and rear faces of the slide in a position in which the front or rear face of the slide **204** faces the Y-direction.

(342) According to some embodiments, the surface of the edge **1130**, the interior edge **1132**, the interior edge of the prong **1120** and/or the prong **1122**, and/or the bottom of the horizontal portion **1112** may have one or more pads (e.g., rubber pads) that may provide compliancy when gripping slides that have varying thicknesses and/or varying label thicknesses. Additionally, the pads may provide additional friction or gripping retention strength. In some embodiments, the pads may be small balls (e.g., rubber balls) inserted into pre-formed or machined cavities conforming to the shape of the balls. Alternatively, the pads may be rubber material that is adhered to the surface of the gripper fingers **1102** and **1104** using a suitable adhesive material.

(343) FIGS. **12A-12B** illustrate aspects of a method for operating an automated histology storage system according to some embodiments.

(344) FIG. **12A** illustrates an example process for an automated histology storage system to receive slides from a user or technician. Although various operations are illustrated in FIG. **12A**, embodiments according to the present disclosure are not limited to the particular operations illustrated in FIG. **12A**. For example, according to some embodiments, there may be additional operations or fewer operations without departing from the spirit and scope of embodiments according to the present disclosure. Additionally, according to some embodiments, the order of operations may vary, unless otherwise stated or implied, without departing from the spirit and scope of embodiments according to the present disclosure.

(345) The operations illustrated in FIG. **12A** may be initiated and/or controlled by various elements of the automated histology storage system **602** including, for example, the control system **650**, the mechanical arm **620**, etc. Unless otherwise stated, any suitable components within the automated histology storage system **602** may be utilized to execute the operations described in FIG. **12A**, and for convenience of description, the various components of the automated histology storage system **602** that perform the operations of the process illustrated in FIG. **12A** may be collectively referred to as “the system.”

(346) As illustrated in FIG. 12A, the process starts and then, at operation **1200**, the system receives patient and/or slide information about slides to be received and/or stored by the automated histology storage system **602**. For example, in some instances, a user or technician may desire to store individual slides, once processed or during the manufacturing process (e.g., in between staining operations, etc.), in the automated histology storage system **602**. The user may, using a user interface (e.g., a graphical user interface displayed on the display panel **614** and one or more input devices) may provide information about a patient and individual slides. The information may be input manually using the user interface, or in some embodiments, an imaging sensor or barcode scanner may be configured to scan unique identifying information (e.g., in the form of a barcode, QR code, characters or numbers, etc.) that may be, for example, affixed to slides and/or a tray or rack. In some embodiments, identifying information may be retrieved from a database (e.g., the database **662**) in response to receiving the unique identifying information. For example, certain patient data or slide data may be stored in the database, and in response to scanning a code or other data corresponding to the slides and/or a tray or rack, the system may retrieve the patient or slide data from the database and match it to the slides and/or the tray/rack.

(347) Additionally, at operation **1200**, the user may provide further information about the type of storage mechanism being used. For example, using the user interface, the user may indicate whether the slides are currently in a tray or rack, and also what the user intends for the automated histology storage system **602** to do with the slides.

(348) At operation **1204**, the system may identify a portal location for the slides to be inserted into the automated histology storage system **602**. For example, in some embodiments, the system may display, on the user interface, information about which portal and/or portal slot the user should deposit the slides. Alternatively, or additionally, the system may display, for example, at the front of one of the portal slots, a light indicating to the user that the adjacent portal slot is where the user should deposit the slides.

(349) At operation **1208**, the system may insert an empty tray or rack carriage into the identified portal slot. According to some embodiments, the empty tray or rack may be selected according to the user input. For example, depending on which type of tray or rack is to be inserted, the system may select a corresponding carriage that has an interior cavity that corresponds to the shape of the tray or rack. The user may then insert the slide tray or rack into the empty carriage.

(350) In some embodiments, the user may push the carriage back into the portal. In some embodiments, one or more sensors (e.g., mechanical buttons, lasers, cameras, pressure sensors, magnets, etc.) may identify the presence of the carriage with the tray or rack successfully inserted. Then, at operation **1212**, the system may retrieve the carriage from the portal from the back side (interior side) of the portal. In some embodiments, the carriage may be transferred (e.g., by the mechanical arm **620**) without human involvement to a nest.

(351) At operation **1216**, the system may retrieve individual slides from the carriage in the nest and, at operation **1220**, the system may transfer the individual slides to a storage location. The retrieval and transferring operations **1216** and **1220** may be executed, for example, by the mechanical arm **620**, using any suitable gripper or gripping operation as described above. For example, according to some embodiments, the retrieval and transferring operations **1216** and **1220** may utilize one or more operations as described above with respect to FIGS. 3J, 4M, and 5J. For example, in some embodiments, the slides may be transferred individually from the carriage to a carrier cassette (e.g., a carrier cassette **500**), and the carrier cassette may then be transferred to a slide carrier queue **530** for storage. When removing slides from the carriage, some embodiments may utilize one or more sensors (e.g., the slide position detection sensor) to identify whether or not a slide is present at any given slide position in a tray or rack, prior to attempting to pick up a slide, for example, as described above with respect to FIGS. 8D and 8E.

(352) Once each of the slides are removed from the carriage and transferred to a storage location, at operation **1224**, the system may return the carriage and the empty tray or rack to an empty portal

slot.

(353) Additionally, according to some embodiments, at or between various operations between **1208** and **1224**, the system may cause the mechanical arm **620** to change grippers in order to perform a subsequent operation. For example, according to some embodiments, the system may utilize a first gripper (e.g., a gripper described with respect to FIGS. **10A-10E**) to handle or manipulate a tray, rack, or carriage (e.g., in the process of removing a carriage from a portal slot, transferring a carriage to a carriage nest, etc.). Then, the system may cause the mechanical arm **620** to switch to a different gripper (e.g., the gripper described with respect to FIG. **4A-4L**, **9A-9E**, or **11A-11B**) in order to pick up or manipulate individual slides, according to the design of the corresponding slide tray or slide rack.

(354) Each of the operations **1200** through **1224** involving the movement, transferring, or storage of slides, carriages, trays, or racks may be executed automatically and without human involvement, other than that user providing input data about the patient or slides, and the type of rack or tray, and the user inserting the rack or tray into a portal.

(355) FIG. **12B** illustrates an example process for an automated histology storage system to retrieve slides from storage and transfer them to a user or technician. Although various operations are illustrated in FIG. **12B**, embodiments according to the present disclosure are not limited to the particular operations illustrated in FIG. **12B**. For example, according to some embodiments, there may be additional operations or fewer operations without departing from the spirit and scope of embodiments according to the present disclosure. Additionally, according to some embodiments, the order of operations may vary, unless otherwise stated or implied, without departing from the spirit and scope of embodiments according to the present disclosure.

(356) The operations illustrated in FIG. **12B** may be initiated and/or controlled by various elements of the automated histology storage system **602** including, for example, the control system **650**, the mechanical arm **620**, etc. Unless otherwise stated, any suitable components within the automated histology storage system **602** may be utilized to execute the operations described in FIG. **12B**, and for convenience of description, the various components of the automated histology storage system **602** that perform the operations of the process illustrated in FIG. **12B** may be collectively referred to as “the system.”

(357) As illustrated in FIG. **12B**, the process starts and then, at operation **1240**, the system receives patient and/or slide information about slides that are to be retrieved from storage by the automated histology storage system **602**. For example, a user may desire for a slide or a group of slides be retrieved and delivered to the user for further processing or to transfer to a new step in the pathology process. The user may search, for example, using a user interface (including, for example, the display panel **614**, and one or more input devices), for a patient, slide, or group of slides, to see if the corresponding slides are currently in storage by the automated histology storage system **602**. The user may then, using the user interface, provide input to request certain slides be returned to the user.

(358) At operation **1244**, the system may identify a portal location or slot at which the slides are to be returned by the automated histology storage system **602**. According to some embodiments, the system may display, on the user interface, information about which portal or portal slot the user should expect the slides. Additionally, or alternatively, the system may display a graphical indicator (e.g., a light) at the portal slot in order to direct the user to the proper portal slot.

(359) According to some embodiments, at operation **1246**, the identified portal slot may not include an empty tray or rack. Accordingly, in some embodiments, the system may prompt the user to insert a rack or tray into a carriage of the tray or rack portal slot. In some embodiments, once the empty tray or rack is inserted into the carriage of the tray or rack portal slot, the system may, using one or more sensors (e.g., mechanical buttons, switches, magnets, lasers, cameras, springs, etc.), identify that the empty tray or rack is inserted. In some embodiments, the user may be prompted to push the empty tray or rack and the carriage into the portal.

(360) At operation **1250**, the system may remove the empty tray or rack and/or the carriage, from the portal from the back or interior side of the portal. At operation **1254**, the system may transfer the tray or rack and/or the carriage to a nest.

(361) Then, at operation **1258**, the system may, based on the user input and requested slides, retrieve each of the requested slides from storage. For example, the slides may be stored in one or more carrier cassettes (e.g., carrier cassettes **500**) and in one or more slide carrier queues (e.g., slide carrier queues). The system may remove the slides, either individually, or as a group with an entire carrier cassette, and transfer each individual slide to the empty tray or rack in the nest. That is, the gripper of the mechanical arm **620** may pick up each slide and transfer it to a corresponding slide position in the tray or rack.

(362) At operation **1262**, the system may pick up the tray or rack and/or the corresponding carriage, and transfer it to an empty portal slot to be retrieved by the user.

(363) The electronic or electric devices and/or any other relevant devices or components according to embodiments of the present invention described herein may be implemented utilizing any suitable hardware, firmware (e.g. an application-specific integrated circuit), software, or a combination of software, firmware, and hardware. For example, the various components of these devices may be formed on one integrated circuit (IC) chip or on separate IC chips. Further, the various components of these devices may be implemented on a flexible printed circuit film, a tape carrier package (TCP), a printed circuit board (PCB), or formed on one substrate. Further, the various components of these devices may be a process or thread, running on one or more processors, in one or more computing devices, executing computer program instructions and interacting with other system components for performing the various functionalities described herein. The computer program instructions are stored in a memory which may be implemented in a computing device using a standard memory device, such as, for example, a random access memory (RAM). The computer program instructions may also be stored in other non-transitory computer readable media such as, for example, a CD-ROM, flash drive, or the like. Also, a person of skill in the art should recognize that the functionality of various computing devices may be combined or integrated into a single computing device, or the functionality of a particular computing device may be distributed across one or more other computing devices without departing from the spirit and scope of the exemplary embodiments of the present invention.

(364) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the present invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and/or the present specification, and should not be interpreted in an idealized or overly formal sense, unless expressly so defined herein.

(365) Although aspects of some example embodiments of a pathology slide manufacturing system and method have been described and illustrated herein, various modifications and variations may be implemented, as would be understood by a person having ordinary skill in the art, without departing from the spirit and scope of embodiments according to the present disclosure.

Accordingly, it is to be understood that a pathology slide manufacturing system and method according to the principles of the present disclosure may be embodiment other than as specifically described herein. The disclosure is also defined in the following claims, and equivalents thereof.

Claims

1. A slide processing system, comprising: a first slide holder configured to hold a plurality of first slides; a second slide holder configured to hold a plurality of second slides; a first carriage having a first interior cavity, the first interior cavity keyed to an exterior side surface around the first slide holder, the first carriage having a contoured surface corresponding to a first gripping location; a

second carriage having a second interior cavity, the second interior cavity keyed to an exterior side surface around the second slide holder, wherein a form factor of the first interior cavity is different from a form factor of the second interior cavity, the second carriage having a contoured surface corresponding to a second gripping location; and a gripper configured to grip the first carriage at the first gripping location and to grip the second carriage at the second gripping location, wherein a contour and location of the first gripping location on the first carriage have a same form factor as a contour and location of the second gripping location on the second carriage.

2. The slide processing system of claim 1, wherein an exterior form factor of the first carriage matches an exterior form factor of the second carriage.

3. The slide processing system of claim 2, wherein the exterior form factor of the first carriage comprises the first gripping location and a plurality of other first gripping locations, and the exterior factor of the second carriage comprises the second gripping location and a plurality of other second gripping locations.

4. The slide processing system of claim 3, wherein the form factor of the first gripping location and a form factor of the plurality of other first gripping locations match the form factor of the second gripping location and a form factor of the plurality of other second gripping locations, respectively.

5. The slide processing system of claim 1, wherein the first slide holder is a first slide tray configured to hold a plurality of first slides arranged in a planar orientation in which front faces of the first slides face toward an upward direction and the first slides are laterally spaced apart from each other from a perspective facing toward the front faces of the first slides, and the second slide holder is a second slide tray configured to hold a plurality of second slides arranged in a planar orientation in which front faces of the second slides face toward an upward direction and the second slides are laterally spaced apart from each other from a perspective facing toward the front faces of the second slides, wherein an exterior form factor of the first slide tray is different from an exterior form factor of the second slide tray.

6. The slide processing system of claim 1, wherein the first slide holder is a first slide rack configured to hold a plurality of first slides arranged in a vertical orientation in which front faces of the first slides face in a first same direction and overlap along the first same direction, and the second slide holder is a second slide rack configured to hold a plurality of second slides arranged in a vertical orientation in which front faces of the second slides face in a second same direction and overlap along the second same direction, wherein an exterior form factor of the first slide rack is different from an exterior form factor of the second slide rack.
